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FEM2Transformer: A Frequency-Domain Enhanced Multi-Scale Multi-Strategy Transformer for Hyperspectral Multispectral Image Fusion

Shuaiqi Liu, Zezheng Zhang, Ruixia Cai, Qi Hu, Bing Li, Cong Liu, and Siyuan Liu

***Abstract*—Although deep learning-based HMIF methods have achieved significant progress, most existing methods employ transformers with single-shape and single-scale windows for long-range dependency modeling, which still suffer from certain limitations in representing complex spatial structures and cross-scale features. In addition, existing Hyperspectral multispectral image fusion (HMIF) methods based on frequency-domain feature learning usually perform frequency-domain analysis solely within the spatial dimension, while ignoring the frequency-domain information embedded in the spectral dimension. To address the above limitations, we design a frequency-domain enhanced multi-scale multi-strategy transformer (FEM2Transformer) for HMIF tasks. Specifically, according to the characteristics of low-resolution hyperspectral images (HSI) and high-resolution multispectral images (MSI), a hyperspectral image frequency-domain enhanced multi-scale mining module and a multispectral image frequency-domain enhanced multi-scale mining module are designed to achieve fine-grained feature extraction for HSI and MSI, respectively. In addition, a dual-domain integration module is proposed to effectively integrate spectral and spatial information within HSI or MSI source images, thereby fully exploring and combining the spatial-domain and spectral-domain features of HSI and MSI. Furthermore, to overcome the limitations of single-shape and single-scale window transformers in modeling complex spatial structures and cross-scale feature representation, a multi-scale multi-strategy window transformer is designed. Meanwhile, a frequency-domain enhanced feature interaction module is constructed to facilitate multimodal feature interaction, capture and enhance the global information of feature maps, and achieve better fusion performance. The experiments carried out on three simulated datasets, including CAVE, Chikusei, and Houston, as well as real-world dataset WorldView-3, verify the presented framework surpasses cutting-edge approaches with respect to numerical indicators along with visual quality across various scenarios. The code is available at https://github.com/cvmdsp/FEM2Transformer.**

***Index Terms*—Hyperspectral Image Fusion, Deep Learning, Frequency-Domain Enhancement, Transformer**

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I. Introduction

HYPERSPECTRAL images (HSIs), which contain abundant and continuous spectrum-based signatures, have exhibited outstanding potential in a variety of computer vision applications, for instance, terrain categorization [1], target identification [2], variation monitoring [3], and target tracing [4]. Nevertheless, owing to the intrinsic limitations of imaging equipment, a fundamental compromise generally exists between spectral resolution and spatial resolution. Although HSIs usually possess dozens or even hundreds of spectral bands, their spatial resolution remains relatively limited, which significantly constrains their effectiveness in fine-grained visual analysis tasks. Therefore, the generation of high-resolution hyperspectral images (HR-HSIs) has become one of the critical problems that urgently requires further investigation in this field [5]. To mitigate this issue, researchers commonly employ a multi-source image fusion framework, in which low-resolution hyperspectral images (LR-HSIs) captured by hyperspectral sensors are integrated with high-resolution multispectral images (HR-MSIs) to generate HR-HSIs with simultaneously enhanced spatial and spectral resolutions [6]. The reconstructed HR-HSIs can further improve the effectiveness and reliability of downstream visual tasks.

In the domain of hyperspectral and multispectral image fusion (HMIF), a host of approaches have been developed by researchers. Current HMIF approaches are mainly categorized into two primary groups, i.e., traditional model-driven techniques and deep learning-based schemes [7]. Conventional techniques primarily depend on carefully designed mathematical formulations and prior knowledge constraints. Typical representative methods include fusion techniques derived from pansharpening theory [8], techniques constructed under matrix decomposition frameworks [9], and techniques established on tensor decomposition models [10]. In recent years, several conventional HMIF techniques have gradually moved beyond explicit pansharpening injection strategies or direct matrix/tensor decomposition formulations. Instead, motivated by the underlying principles of these techniques, researchers have developed new traditional fusion frameworks with enhanced reconstruction capability. For instance, Long et al. [11] introduced an HMIF algorithm based on joint spatial-spectral optimization. By establishing degradation observation models for HSI and MSI acquisition processes, the method simultaneously exploited spatial and spectral characteristics under an integrated optimization scheme, thereby achieving HR-HSI

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reconstruction even under large upsampling factors. Cao et al. [12] proposed a kernel-free, semi-blind HSI-MSI fusion framework formulated as a discrete–continuous maximum a posteriori estimation problem. By introducing a categorical prior based on truncated angular similarity to construct a data-dependent prior center for each pixel, the method implicitly absorbs unknown spatial degradation and mitigates mismatches caused by cross-sensor differences, thereby improving the robustness of model-driven fusion in practical imaging scenarios.

Compared with traditional algorithms, deep learning-driven HMIF algorithms effectively break through the constraints imposed by handcrafted prior knowledge. Existing deep learning-oriented fusion algorithms are commonly classified into two representative branches, namely convolutional neural network (CNN)-driven methods and transformer-based algorithms. For instance, Liu et al. [13] constructed an HMIF method with an approximate spectral mapping strategy, which enhanced the capability of the network in recovering fine-grained details. Furthermore, they introduced an adaptive defect-aware optimization loss to areas exhibiting obvious spectral discrepancies, thus enhancing the consistency between the fusion results and reference images. Mu et al. [14] presented an HMIF framework utilizing a mask-guided cross-attention architecture, which efficiently captured spatial and spectral representations by incorporating a CNN-oriented mask-based feature extraction component, a cross-feature interaction module integrating channel-wise and spatial attention, and a progressive multi-scale reconstruction strategy. Although CNN-based methods have achieved remarkable success, particularly in modeling local spatial structures, their inherently limited receptive ranges make it challenging to capture long-distance feature correlations, which may lead to fusion outputs lacking sufficient global contextual representation [15]. Recently, transformer architectures have exhibited greater advantages in complicated multi-source image fusion problems because they are capable of establishing global spatial-spectral correlations while effectively learning long-range feature interactions. For example, Yu et al. [16] introduced an HMIF framework built upon a transformer-guided interaction U-Net. The proposed framework enabled adequate cross-modal feature communication by employing stacked cross-modal representation learning modules together with transformer-based cross-modal fusion gates. Meanwhile, the bottleneck structure within the U-Net architecture efficiently alleviated the computational overhead introduced by the transformer, thereby achieving superior spatial detail preservation and spectral consistency in HMIF applications. Lee et al. [17] devised an HMIF framework using a dual-interaction network. Through the design of mutually reinforced HR-MSI and LR-HSI subnetworks, effective extraction and reutilization of cross-domain representations are achieved. As a result, the fusion performance of the model was significantly improved while maintaining model compactness. Although transformers overcome the limitation of CNNs in terms of restricted receptive fields and are capable of extracting global contextual information, thereby achieving competitive fusion results in the HMIF field, their computational complexity increases quadratically with image size. Although some window-based transformer variants have been proposed to improve computational efficiency, most existing methods are still based on single-window partition strategies and fixed window scales for feature modeling. This still leads to certain limitations in representing complex spatial structures and cross-scale features, which further restrict the fusion accuracy of HMIF tasks.

Current window-based transformer architectures generally employ fixed-shape and fixed-scale windows, which limits their capability in representing complicated spatial patterns and capturing cross-scale information. Moreover, most existing HMIF algorithms utilizing frequency-domain feature learning mainly conduct frequency decomposition along the spatial dimension, while insufficiently exploring the frequency characteristics embedded in the spectral dimension. To address such limitations, we propose FEM2Transformer, a frequency-domain enhanced multi-scale multi-strategy transformer (FEM2Transformer), for HMIF, which consists of four components: the hyperspectral image frequency-domain enhanced multi-scale mining (HSI-FEMM) module, the multispectral image frequency-domain enhanced multi-scale mining (MSI-FEMM) module, the dual-domain integration (DDI) module, and the frequency-domain enhanced feature interaction (FEFI) module. The detailed architectures and functionalities of these modules are presented in Section III, and the main contributions of this paper are summarized as follows:

(1) We develop FEM2Transformer to achieve fine-grained feature extraction and effective interactive fusion from source images. By incorporating frequency-domain representations from both the spatial and spectral domains and adopting a multi-scale multi-strategy window transformer, the proposed framework strengthens the capability to characterize complex spatial structures and cross-scale information.

(2) We propose the HSI-FEMM and MSI-FEMM modules to perform fine-grained frequency-domain feature modeling for HSI and MSI, respectively, from a spatial-spectral dual-domain collaborative perspective. In addition, the DDI module is introduced to strengthen and fuse the spectral and structural representations of the two modalities, providing more comprehensive feature representations for subsequent fusion.

(3) We propose a multi-scale and multi-strategy window transformer. By introducing a frequency-domain enhancement mechanism, the FEFI module jointly models features using different window shapes and scales and adopts frequency-domain representations to strengthen the characterization of geospatial and spectral attributes, thereby further enhancing the comprehensive fusion capability of the proposed method.

The follow-up parts of this work are laid out as listed below. Section II introduces the related studies for HMIF. Section III provides a introduction to the proposed FEM2Transformer framework. Section IV presents and discusses the experiment results from simulated datasets. Section V reports the experimental outcomes obtained on practical real-scene datasets. Finally, Section VI gives the conclusion of this work.

II. Related Work

A. Transformer-Based HMIF Methods

The transformer was initially built for machine translation applications. With the continuous advancement of related

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research, it has been progressively extended to computer vision tasks and has achieved success. Owing to the capability of self-attention mechanisms in capturing long-distance correlations among image regions, Transformers have exhibited powerful feature learning ability in a wide range of visual applications. Researchers have further incorporated Transformer architectures into image fusion research to effectively enhance the visual effect of reconstructed imagery. For instance, Ma et al. [18] put forward a dual cross-fusion framework built upon Transformer architecture. By virtue of adopting a multi-head cross-attention strategy, the proposed algorithm enables effective feature exchange between HSI and MSI modalities. Following cross-modal feature transmission, a Swin Transformer component is additionally introduced to strengthen contextual self-attention learning, thereby significantly improving the fusion performance of the HMIF model. However, the standard transformer suffers from high computational complexity. To alleviate this problem, numerous transformer variants have been developed. For example, Wang et al. [19] proposed a gradual multi-scale axial attention fusion framework, which integrates multi-scale convolution and axial attention mechanisms. This design effectively reduces computational complexity while enhancing global feature modeling capability. The method models long-range dependencies along spatial and spectral dimensions using axial attention and achieves collaborative optimization of hyperspectral and multispectral data in both spatial and spectral dimensions through progressive feature interaction and hierarchical fusion strategies, significantly improving HR-HSI reconstruction quality. Fu et al. [20] put forward a multi-scale space-spectrum dilated Transformer architecture for HMIF. By designing a dual-branch architecture composed of a spatial dilated multi-scale Transformer and a spectral dilated positional embedding Transformer, the method alleviates the limitations of CNNs in local modeling and the high computational cost of standard Transformers, while enabling efficient modeling of HSI spatial-spectral features. Liu et al. [21] presented a dual-phase Transformer-based fusion architecture established upon a multi-head cross-attention mechanism with overlapping rectangular windows. A sequential feature interaction strategy is adopted, where spectral information of HSI is first injected into MSI features, and then spatial information of HSI is fed back into MSI features. This design effectively alleviates feature interference caused by simultaneous interaction and achieves a more sufficient spectral and spatial collaborative fusion. Jia et al. [22] presented a multi-scale spatial-spectral transformer network for HMIF. It adopts a two-branch architecture with cross-modality concatenation and exploits a spectral Transformer and spatial Transformers with multi-scale band/patch embeddings to extract spectral features from HSI and spatial features from MSI, respectively. The method effectively captures global spatial-spectral dependencies for HSI-MSI fusion. Xiao et al. [23] proposed a region-aware mixture-of-experts (MoE) network for HMIF. By designing an attention-sharing module to propagate attention across modalities and a region-aware MoE composed of multiple region-specific experts together with an invariant-weight expert, the method achieves more balanced cross-modal interaction and captures heterogeneous spatial-spectral patterns across different regions.

Although the aforementioned transformer-based HMIF methods have improved computational efficiency through various transformer variants, these models still suffer from the limitation of using single-shape and single-scale windows, making it difficult to fully represent complex spatial structural information. Therefore, we construct a multi-scale and multi-strategy window transformer and introduce the FEFI module to enable sufficient cross-modal interaction, thereby enhancing the capability of modeling complex spatial structures.

B. Frequency-Domain Feature Learning-Based HMIF Methods

Compared with the spatial domain, the frequency domain describes the structural characteristics and detailed distribution of images from a frequency-analysis perspective. Specifically, high-frequency components mainly correspond to edges, textures, and repetitive structures, whereas low-frequency components reflect basic contours and scale-related variations. When frequency-domain representations are integrated with spatial-domain features, the model can obtain a more comprehensive description of structural and detailed information. Consequently, more researchers have devoted attention to frequency-domain learning and introduced it into HMIF applications. For instance, Tan et al. [24] proposed an HMIF framework based on space-frequency and geospatial characteristic learning. This method addresses the limitation of CNNs in global modeling by jointly modeling spectral frequency and spatial attributes. It contains a dimension feature extraction stream together with a space feature extraction stream, and an inter-domain feature fusion component is designed to facilitate complementary interaction across the two categories of features, thereby more effectively modeling the correspondence between MSI and HSI. Zhu et al. [25] introduced an HMIF framework driven by the classic Kolmogorov–Arnold network (KAN) dual spatial-frequency domain implicit guided sampling-oriented strategy. By integrating the strengths of KAN and implicit neural representation learning, the proposed method performs guided sampling on HSI data across spatial and frequency fields, enabling efficient integration of spatial and spectral characteristics. Xu et al. [26] devised an enhanced spatial–frequency collaborative HMIF method. By introducing a selective frequency decomposition module, features are modeled in the frequency domain using a global filter that incorporates both phase and amplitude information. This approach effectively captures global dependencies from the frequency perspective while lowering the computational pressure of transformer-based global self-attention mechanisms and significantly improving fusion performance. Tan et al. [27] addressed the limitation of global Fourier-based modeling by proposing a hierarchical frequency fusion network. This method decomposes features of PAN and MSI into spatial information, global frequency information, and local frequency information, and performs independent fusion at different levels followed by hierarchical integration. This effectively enhances the model's capability in hierarchical modeling of spatial- and frequency-domain representations, thereby substantially enhancing fusion performance in pansharpening tasks.

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Although the aforementioned HMIF methods based on frequency-domain feature learning have achieved promising fusion performance, most of these models mainly perform frequency-domain modeling and analysis only in the spatial dimension, without fully exploring the spectral structural information contained in the spectral dimension. Therefore, we extend frequency-domain analysis from the spatial domain to the spectral domain for spatial-spectral dual-domain collaboration, and design the MSI-FEMM and HSI-FEMM modules together with the DDI module to strengthen frequency-domain feature representation.

III. Proposed Method

A. The Network Framework

The FEM2Transformer mainly consists of four components: the HSI-FEMM module, the MSI-FEMM module, the DDI module, and the FEFI module. The complete network framework is illustrated in Fig. 1. First, FEM2Transformer performs shallow feature extraction on the source images through convolution layers and upsampling operations. The specific process is described as follows:

$$F_m^0 = C_{onv}(X), F_h^0 = C_{onv}(Up(Y)) \tag{1}$$

where $X \in \mathbb{R}^{H\times W\times c}$ represents the HR-MSI, $Y \in \mathbb{R}^{h\times w\times C}$ represents the LR-HSI, $Up(\cdot)$ represents bilinear interpolated upsampling operation, $C_{onv}(\cdot)$ represents convolution operation with kernel dimension of 3×3, $F_m^0 \in \mathbb{R}^{H\times W\times B}$ indicates the primitive feature map belonging to MSI, and $F_h^0 \in \mathbb{R}^{H\times W\times 2B}$ indicates the primitive feature map derived from HSI. Since HSI contains richer spectral information while MSI contains richer spatial information, FEM2Transformer feeds the two shallow features into the HSI-FEMM module and the MSI-FEMM module, respectively, to finely extract the global and local features of HSI and MSI according to their individual characteristics. The concrete implementation workflow is demonstrated as follows:

$$F_m^1 = MF_{EMM}(F_m^0), F_h^1 = HF_{EMM}(F_h^0) \tag{2}$$

where $MF_{EMM}(\cdot)$ indicates the feature mapping operation of the MSI-FEMM module, $HF_{EMM}(\cdot)$ represents the mapping function of the HSI-FEMM module, and $F_m^1 \in \mathbb{R}^{H\times W\times B}$ and $F_h^1 \in \mathbb{R}^{H\times W\times B}$ denote the outputs of the MSI-FEMM and HSI-FEMM modules, correspondingly. Although HSI contains rich spectral information, it still preserves certain spatial information, and the same applies to MSI. Therefore, FEM2Transformer transmits the dual-modal feature representations yielded by HSI-FEMM and MSI-FEMM components into the DDI module, respectively, to extract and integrate the dual-domain spatial-spectral feature information contained in both HSI and MSI. The above workflow can be expressed as follows:

$$F_m^2 = DDI(F_m^1), F_h^2 = DDI(F_h^1) \tag{3}$$

where $DDI(\cdot)$ represents the mapping function of the DDI module, $F_m^2 \in \mathbb{R}^{H\times W\times B}$ and $F_h^2 \in \mathbb{R}^{H\times W\times B}$ denote the second-layer feature maps of MSI and HSI, respectively. At this stage, the output feature maps of the two individual branches have already contained relatively rich and comprehensive features. They are then simultaneously fed into the FEFI module to achieve sufficient interaction and complementary enhancement between the two modalities, which can be expressed as follows:

$$F_{mh} = FEFI(F_m^2, F_h^2) \tag{4}$$

where $FEFI(\cdot)$ represents the mapping function of the FEFI module, and $F_{mh} \in \mathbb{R}^{H\times W\times B}$ denotes the unified feature map derived from HSI and MSI modalities. Finally, the interacted feature map is passed through convolution layers and activation functions to complete image fusion. Meanwhile, to further aggregate contextual information, FEM2Transformer employs skip linkage to superpose upsampled LR-HSI and integrated feature maps, which acquire HR-HSI that possesses superior spatial resolution and comprehensive spectral information. The specific workflow is elaborated as follows:

$$Z = C_{onv}(L_{relu}(C_{onv}(F_{mh}))) + UP(Y) \tag{5}$$

where $L_{relu}(\cdot)$ represents the LeakyReLU activation function, and $Z \in \mathbb{R}^{H\times W\times C}$ denotes the estimated HR-HSI generated by the model. The internal architectures of these modules will be elaborated in subsequent sections.

B. Hyperspectral Image Frequency-Domain Enhanced Multi-Scale Mining Module

The HSI-FEMM module is composed of a spectral multi-scale perception (SMP) module, a multi-scale aligned mixing (MSAX) module, and a frequency-domain channel enhancement (FDCE) module, as illustrated in Fig. 1 and Fig. 2. The SMP module extracts multi-scale spectral representations through grouped convolutions; the MSAX module captures multi-scale spatial representations; and the FDCE module enhances spectral frequency-domain features via DCT-based channel attention. The detailed structures and computations of these three modules are presented below. The specific process can be described as follows:

$$F_h^1 = M_{SAX}(S_{MP}(F_h^0)) + F_{DCE}(S_{MP}(F_h^0)) \tag{6}$$

where $S_{MP}(\cdot)$ stands for the mapping function of the SMP module, $M_{SAX}(\cdot)$ represents the mapping function of the MSAX module, $F_{DCE}(\cdot)$ represents the mapping function of the FDCE module, and $F_h^1 \in \mathbb{R}^{H\times W\times B}$ denotes the output feature map of the HSI-FEMM module. Detailed illustrations of the three modules are presented in the following parts.

The SMP module performs grouped processing of feature maps at three different granularities along the spectral dimension. Specifically, the spectral channels are divided into 4, 8, and 16 groups, respectively, and shared convolution kernels are applied within each group for feature extraction. This multi-scale grouping strategy enables the model to learn spectral representations across multiple resolutions from coarse to fine, effectively enhancing the capability of the model in characterizing spectral information while improving representation precision. In addition, it increases the model's sensitivity to

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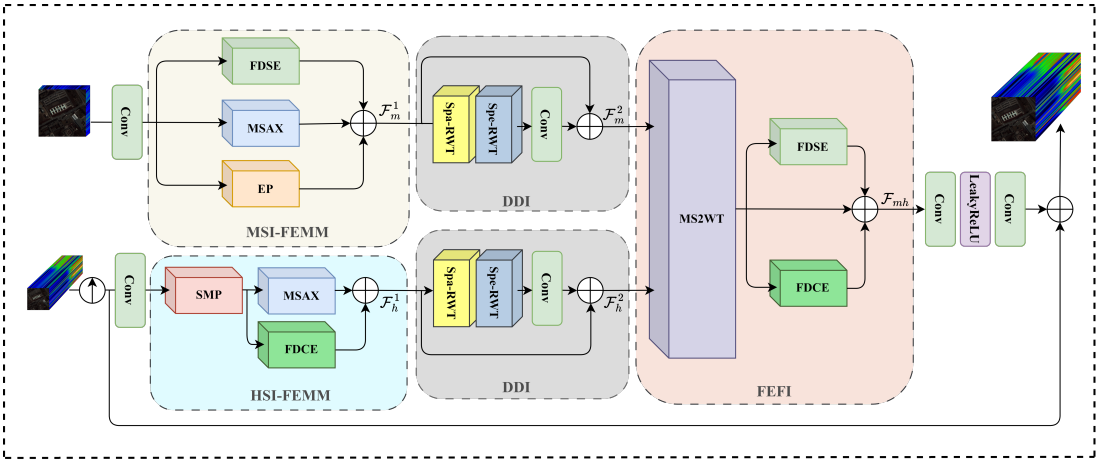


Fig. 1. Overall structure of the FEM2Transformer. The FEM2Transformer can be divided into four components: the HSI-FEMM module, the MSI-FEMM module, the DDI module, and the FEFI module.

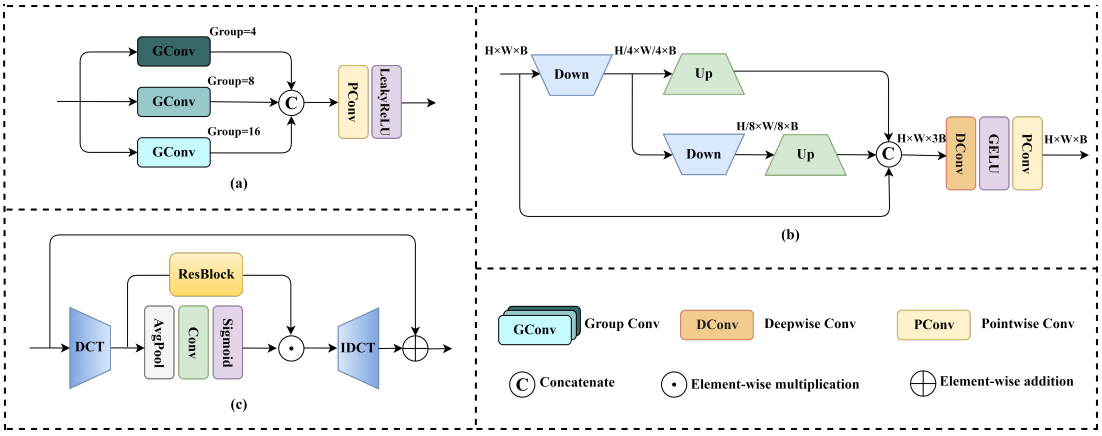


Fig. 2. Overall structure of major components belonging to the HSI-FEMM module. (a) Network structure of the SMP module, (b) Network structure of the MSAX module, (c) Network structure of the FDCE module.

spectral variations in complex scenes. The detailed computation process is as follows:

$$F_{h0}^{1i} = GC_{onvi}(F_h^0), \quad i = 0, 1, 2 \tag{7}$$

$$F_{h0}^{1} = L_{relu}(PC_{onv}(C_{at}(F_{h0}^{10}, F_{h0}^{11}, F_{h0}^{12}))) \tag{8}$$

where $GC_{onv_i}(\cdot)$ denotes the grouped convolution operation with the number of groups set to i (i.e., 4, 8, or 16), $C_{at}(\cdot)$ represents the concatenation function, $PC_{onv}(\cdot)$ represents the point-wise convolution, and $L_{relu}(\cdot)$ denotes the LeakyReLU activation function. $F_{h0}^{1i} \in \mathbb{R}^{H\times W\times B}$ indicates the output feature map following grouped convolution calculation, and $F_{h0}^{1} \in \mathbb{R}^{H\times W\times B}$ denotes the resultant feature representation of the SMP module.

The MSAX module constructs feature representations at three different spatial resolutions through progressive downsampling operations, to capture spatial structural information at multiple scales within the image. Subsequently, the MSAX module upsamples the low-resolution feature maps back to a unified scale and fuses them with the high-resolution features, thereby achieving effective extraction and alignment of multi-scale spatial features. Such a strategy not only strengthens the network's capability to jointly perceive local fine features and global holistic structures, but also improves its robustness and representational capacity in handling complex spatial variations. The detailed process is as follows:

$$F_{h1}^{10} = D_{own}(F_{h0}^{1}), F_{h1}^{11} = D_{own}(F_{h1}^{10}) \tag{9}$$

$$F_{h1}^{1} = PC_{onv}(G_{elu}(DC_{onv}(C_{at}(U_p(F_{h1}^{10}), U_p(F_{h1}^{11}), F_{h0}^{1})))) \tag{10}$$

where $D_{own}(\cdot)$ denotes the downsampling process, $U_p(\cdot)$ denotes the upsampling process, $DC_{onv}(\cdot)$ indicates the depth-wise conv operation, and $G_{elu}(\cdot)$ denotes the GELU non-linear activation function. $F_{h1}^{10} \in \mathbb{R}^{H/4\times W/4\times B}$ and $F_{h1}^{11} \in \mathbb{R}^{H/8\times W/8\times B}$ stand for feature representations with two distinct scales, and $F_{h1}^{1} \in \mathbb{R}^{H\times W\times B}$ stands for the final output feature yielded by the MSAX module.

The designed FDCE module starts by translating the original spatial feature representations to the frequency domain through DCT. Unlike a learnable spectral transform, DCT introduces no additional learnable parameters as a parameter-free orthogonal transform, which helps avoid overfitting and reduces training instability given the limited number of original HSI scenes. Beyond parameter efficiency, DCT provides a physically interpretable frequency decomposition in which low-frequency components correspond to basic contours and scale information, while high-frequency components capture edges and textures, and this explicit and consistent frequency

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semantics is well aligned with our goal of enhancing structural and detailed representations in a controllable manner. Specifically, a two-dimensional DCT is applied to each channel to explore the frequency structure of features. Considering the high redundancy of HSI in the spectral dimension, the FDCE module performs frequency-domain modeling along the spectral dimension and introduces a channel attention mechanism in the frequency domain. This mechanism dynamically models the importance of different spectral channels, emphasizing key frequency components while suppressing redundant features. Finally, the FDCE module recovers the refined frequency characteristics toward the spatial field via the inverse discrete cosine transform (IDCT) and fuses them with spatial feature maps via element-wise summation, achieving effective integration of frequency and spatial information. The detailed computation process is as follows:

$$F_{h2}^{10} = DCT(F_{h0}^{1}) \tag{11}$$

$$F_{h2}^{11} = S_{ig}(C_{onv}(AP_{ool}(F_{h2}^{10}))) \odot RB(F_{h2}^{10}) \tag{12}$$

$$F_{h2}^{1} = IDCT(F_{h2}^{11}) + F_{h0}^{1} \tag{13}$$

where $DCT(\cdot)$ denotes the DCT function, $RB(\cdot)$ represents the mapping function of the ResBlock, $AP_{ool}(\cdot)$ denotes the average pooling (AvgPool) operation, $S_{ig}(\cdot)$ represents the Sigmoid activation function, $IDCT(\cdot)$ denotes the inverse discrete cosine transform (IDCT), and $F_{h2}^{1} \in \mathbb{R}^{H\times W\times B}$ stands for the final output feature yielded by the FDCE module.

C. Multispectral Image Frequency-Domain Enhanced Multi-Scale Mining Module

The MSI-FEMM module consists of an edge perception (EP) module, an MSAX module, and a frequency-domain spatial enhancement (FDSE) module, as shown in Fig. 1. Each component in the MSI-FEMM module is carefully designed based on the characteristic that MSI contains richer spatial information than spectral information, with the aim of fully exploiting spatial structures and texture features in MSI, thereby providing precise and detailed spatial information for subsequent fusion. The EP module extracts noise-robust and structurally informative edge features using the Canny operator, followed by further refinement to fully capture spatial edges and textures in MSI. Its detailed architecture is shown in Fig. 3(a). The MSAX module plays the same role as in the HSI-FEMM module, capturing multi-scale spatial representations via multi-scale feature learning, thereby enhancing the completeness and reliability of feature representation. The FDSE module shifts original spatial representations to the frequency domain, aiming at feature reinforcement, making up for the missing frequency information of MSI feature data. Its concrete architecture is shown in Fig. 3(b). The specific computation process of the MSI-FEMM module is as follows:

$$F_{m}^{1} = EP(F_{m}^{0}) + M_{SAX}(F_{m}^{0}) + F_{DSE}(F_{m}^{0}) \tag{14}$$

where $EP(\cdot)$ indicates the mapping function of the EP module, $M_{SAX}(\cdot)$ represents the mapping function of the MSAX module, $F_{DSE}(\cdot)$ represents the mapping function of the FDSE module, and $F_{m}^{1} \in \mathbb{R}^{H\times W\times B}$ stands for the final output feature generated by the MSI-FEMM module. Detailed introductions to EP and FDSE modules will be presented later.

The EP module first employs the Canny operator to extract coarse edge features that are both structurally informative and robust to noise, thereby highlighting key spatial contours in MSI. Since these edge features are relatively coarse, the EP module further refines them through multiple ResBlocks and activation functions to perform spatial attention, allowing the model to automatically emphasize important regions while generating enhanced edge representations. The enhanced edge representation is subsequently integrated with the input features through residual connections and further subtracted from the original coarse edge features before refinement. This operation helps suppress redundant or low-quality edge responses. By adopting a joint enhancement-and-suppression strategy, the EP module effectively extracts discriminative edge and texture features in MSI, providing accurate spatial structural information for subsequent fusion tasks. The specific calculation flow of the EP module is presented below:

$$F_{m0}^{10} = C_{anny}(F_{m}^{0}) \tag{15}$$

$$F_{m0}^{11} = S_{ig}(L_{relu}RB^{2}(F_{m0}^{10})) \odot RB(F_{m0}^{10}) \tag{16}$$

$$F_{m0}^{1} = F_{m0}^{11} + F_{m}^{0} - F_{m0}^{10} \tag{17}$$

where $C_{anny}(\cdot)$ denotes the Canny operator, $L_{relu}RB^{2}(\cdot)$ represents the operation of applying a ResBlock followed by a LeakyReLU activation function twice consecutively, $RB(\cdot)$ denotes the identity mapping function of the ResBlock, $S_{ig}(\cdot)$ indicates the Sigmoid activation module, and $\odot$ stands for element-wise product. $F_{m0}^{10} \in \mathbb{R}^{H\times W\times B}$ represents the coarse edge features extracted by the Canny operator, $F_{m0}^{11} \in \mathbb{R}^{H\times W\times B}$ represents the refined edge features, and $F_{m0}^{1} \in \mathbb{R}^{H\times W\times B}$ denotes the final output of the EP module.

The FDSE module follows a design philosophy similar to that of the FDCE module, as it also leverages frequency-domain features to compensate for the limitations of spatial-domain representations, thereby enhancing the structural and detailed expressiveness of feature maps. The key difference is that the FDSE module is tailored to MSI, which contains richer spatial information. It performs feature enhancement in the frequency domain using convolutional layers and residual blocks to further exploit latent structural information embedded in frequency components. The detailed computational steps of the FDSE module are listed below:

$$F_{m1}^{1} = IDCT(RB(L_{relu}(C_{onv}(DCT(F_{m}^{0}))))) + F_{m}^{0} \tag{18}$$

where $DCT(\cdot)$ denotes the DCT, $RB(\cdot)$ represents the mapping function of the ResBlock, $C_{onv}(\cdot)$ denotes the convolution operation, $L_{relu}(\cdot)$ represents the LeakyReLU activation function, $IDCT(\cdot)$ denotes the IDCT, and $F_{m1}^{1} \in \mathbb{R}^{H\times W\times B}$ represents the output feature map of the FDSE module.

Although both the FDCE and FDSE modules rely on the same DCT-based frequency transform, frequency modeling behaves differently when applied along the spectral dimension and along the spatial dimensions, which is also why they are designed as spectral and spatial frequency enhancements, respectively. For the HSI branch, adjacent spectral bands

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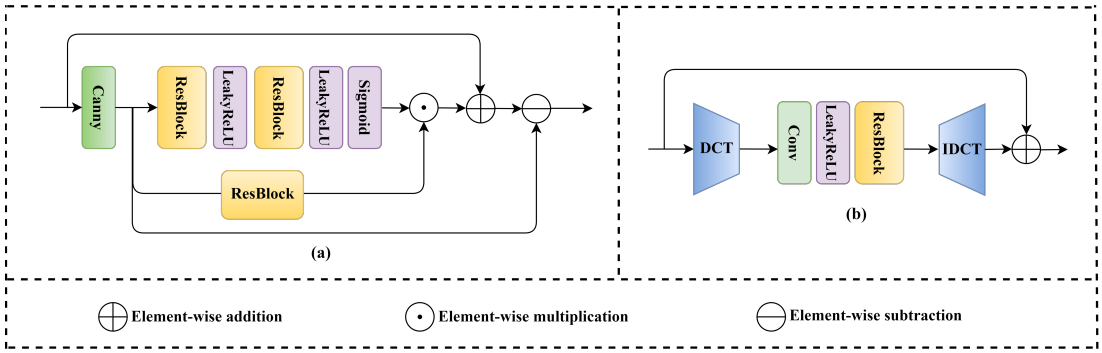


Fig. 3. Overall structure of major components belonging to the MSI-FEMM module. (a) Network structure of the EP module, (b) Network structure of the FDSE module.

are highly correlated, and the spectral signature of each material is intrinsically smooth; consequently, the frequency content along the spectral dimension is strongly low-frequency dominant, so that a small number of low-frequency components already compactly represent the overall spectral shape while the remaining high-frequency components encode fine spectral absorptions. The FDCE module therefore exploits this spectral redundancy through frequency-domain channel attention to enhance the spectral-frequency representation. In contrast, each MSI band contains abundant spatial structures such as edges and textures, whose energy is concentrated in localized high-frequency components; the 2D DCT applied to the spatial dimensions therefore captures structural details that are less correlated across bands. Because the intrinsic statistics of the two domains differ, the same orthogonal DCT yields features of fundamentally different semantic meaning (spectral-frequency versus spatial-frequency), and the FDCE and FDSE modules are accordingly devised to exploit these complementary frequency characteristics.

D. Dual-Domain Integration Module

The DDI module primarily contains two important components: the spatial rectangular window transformer (Spa-RWT) and the spectral rectangular window transformer (Spe-RWT), as shown in Fig. 1. It is well known that HSI contains rich spectral information, while MSI provides higher spatial resolution. Therefore, existing studies often emphasize learning spectral representations from HSI together with spatial representations from MSI for fusion. However, this does not imply that the spatial information in HSI and the spectral information in MSI are negligible. Ignoring these components may lead to incomplete fusion or information loss. To this end, so as to comprehensively exploit and integrate spatial-domain and spectral-domain characteristics from HSI and MSI, the DDI module is introduced at the end of each branch. The DDI module first employs the Spa-RWT module to deeply model and fuse spatial information, followed by the Spe-RWT module to extract and integrate spectral information. Through this cascaded design, the DDI module achieves dual-domain feature integration across both spatial and spectral dimensions, effectively enhancing the complementarity between multimodal features and providing richer and more accurate representations for the final fusion results. The detailed computational steps of the DDI module are listed below:

$$F_{h,m}^{2} = C_{onv}(Spe_{RWT}(Spa_{RWT}(F_{h,m}^{1}))) + F_{h,m}^{1} \tag{19}$$

where $C_{onv}(\cdot)$ denotes the convolution operation, $Spa_{RWT}(\cdot)$ represents the transformation function of the Spa-RWT module, and $Spe_{RWT}(\cdot)$ represents the transformation function of the Spe-RWT module. $F_{h,m}^{1} \in \mathbb{R}^{H\times W\times B}$ denotes the first-layer feature map of HSI or MSI, i.e., F_{h}^{1} or F_{m}^{1}. $F_{h,m}^{2}$ denotes the second-layer feature representation of HSI or MSI, i.e., F_{h}^{2} or F_{m}^{2}. In the following, the network architectures of the Spa-RWT and Spe-RWT modules are described in detail.

The Spa-RWT module mainly extracts and enhances spatial information from HSI or MSI through spatial rectangular window attention (Spa-RWA), as shown in Fig. 4(a). Similar to [28], Spa-RWA employs overlapping window attention to extract global features along the spatial dimension, and its architecture is illustrated in Fig. 4(b). Specifically, Spa-RWA introduces a self-attention mechanism inside each window to capture spatial dependencies by modeling correlations among pixels within an identical window. Due to the adoption of overlapping windows, any pair of spatial positions in the image can be indirectly associated through multiple shared regions. As feature information propagates and merges within these overlapping regions, long-distance dependencies gradually spread throughout the entire image, enabling effective modeling of global features. In this work, $Q \in R^{H\times W\times B}$ represents the query vector, $K \in R^{H\times W\times B}$ represents the corresponding key vector, and $V \in R^{H\times W\times B}$ represents the value vector. Furthermore, $Q_{Rwin} \in R^{num\times {s_1}^2\times B}$ represents the query vector after standard rectangular window partitioning, while $K_{Rwin} \in R^{num\times {s_2}^2\times B}$ and $V_{Rwin} \in R^{num\times {s_2}^2\times B}$ represent the key and value vectors after overlapping rectangular window partitioning, respectively. s_1 represents the scale of the conventional window, s_2 denotes the size of the overlapping window, and num denotes the total number of windows. In the DDI module, we set $s_1 = 4$ and $s_2 = 6$. The window sliding stride is set to 4. The specific computation process of Spa-RWA is as follows:

$$\begin{cases} Q = L_{in}(f), Q_{Rwin} = RS(Q) \\ K = L_{in}(f), K_{Rwin} = RO(K) \\ V = L_{in}(f), V_{Rwin} = RO(V) \end{cases} \tag{20}$$

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$$Att = S_{oftmax}\left(\frac{Q_{Rwin}{K_{Rwin}}^{T}}{\sqrt{d}} + E\right)V_{Rwin} \tag{21}$$

$$f_{att} = L_{in}(R_{eshape}(Att)) \tag{22}$$

where $L_{in}(\cdot)$ denotes the linear layer mapping function, $RS(\cdot)$ represents the standard rectangular window partition operation, $RO(\cdot)$ denotes the overlapping rectangular window partition operation, d represents the dimensionality of the query/key vectors, E denotes the relative positional encoding, $S_{oftmax}(\cdot)$ stands for the Softmax activation function, and $R_{eshape}(\cdot)$ denotes the reshape operation. To further enhance the discriminability of feature representations and the selectivity of inter-channel information, a channel attention mechanism is introduced after the spatial attention operation in Spa-RWA. This mechanism adaptively adjusts the importance distribution across channels from a global perspective, thereby achieving joint spatial–channel feature enhancement and improving the model's response to key regions and informative features. The concrete implementation workflow is formulated as shown below:

$$f_{out} = S_{ig}(AP_{ool}(PC_{onv}(C_{at}(Q, K)))) \odot f_{att} + f_{att} \tag{23}$$

where $C_{at}(\cdot)$ denotes the concatenation function, $PC_{onv}(\cdot)$ denotes the point-wise convolution operation, and $AP_{ool}(\cdot)$ denotes the AvgPool operation. $S_{ig}(\cdot)$ stands for the Sigmoid activation function. $\odot$ represents pointwise multiplication, $f_{out} \in R^{H\times W\times B}$ represents the output feature of Spa-RWA.

The Spe-RWT module extracts and enhances spectral information of HSI or MSI through spectral rectangular window attention (Spe-RWA), as shown in Fig. 4(c). Notably, since the Spe-RWT module focuses more on spectral information, a channel attention block (CAB) from [29] is introduced to replace the conventional multilayer perceptron (MLP). The CAB dynamically distributes weighting coefficients across diverse feature channels, thereby enhancing the network's responsiveness to key spectral channels and further improving the discriminative ability of spectral features. Spe-RWA shares the same overlapping window mechanism as Spa-RWA. The key difference is that Spe-RWA performs attention operations along the spectral dimension, placing greater emphasis on modeling global dependencies among spectral channels, thereby improving spectral feature representation. Its detailed architecture is shown in Fig. 4(d). In this work, $\hat{Q} \in R^{H\times W\times B}$ denotes the query vector, $\hat{K} \in R^{H\times W\times B}$ denotes the key vector, and $\hat{V} \in R^{H\times W\times B}$ denotes the value vector. Furthermore, $\hat{Q}_{Rwin} \in R^{num\times {s_2}^2\times B}$ represents the query vector after overlapping rectangular window partitioning, $\hat{K}_{Rwin} \in R^{num\times {s_2}^2\times B}$ represents the key vector after overlapping rectangular window partitioning, and $\hat{V}_{Rwin} \in R^{num\times B\times {s_1}^2}$ represents the value vector after standard rectangular window partitioning. s_1, s_2 and the stride of window sliding keeps the same parameter setting as the Spa-RWA module. The detailed computational steps of Spe-RWA are listed below:

$$\begin{cases} \hat{Q} = L_{in}(\hat{f}), \hat{Q}_{Rwin} = RO(\hat{Q}) \\ \hat{K} = L_{in}(\hat{f}), \hat{K}_{Rwin} = RO(\hat{K}) \\ \hat{V} = L_{in}(\hat{f}), \hat{V}_{Rwin} = RS(\hat{V}) \end{cases} \tag{24}$$

$$\hat{Att} = S_{oftmax}\left(\frac{\hat{Q}^{T}_{Rwin}\hat{K}_{Rwin}}{\sqrt{d}}\right)\hat{V}_{Rwin} \tag{25}$$

$$\hat{f}_{att} = L_{in}(R_{eshape}(\hat{Att})) \tag{26}$$

where $L_{in}(\cdot)$ denotes the linear layer mapping function, $RS(\cdot)$ represents the standard rectangular window partition operation, $RO(\cdot)$ denotes the overlapping rectangular window partition operation, d represents the dimensionality of the query/key vectors, E denotes the relative positional encoding, $S_{oftmax}(\cdot)$ indicates the Softmax activation module, and $R_{eshape}(\cdot)$ denotes the reshape operation. $\hat{f}$ represents the input feature map of Spe-RWA, and $\hat{f}_{att} \in R^{H\times W\times B}$ represents the output feature representation Spe-RWA. After channel modeling, Spe-RWT further introduces a spatial attention mechanism, which adaptively adjusts features according to the semantic importance of different spatial locations, facilitating the model's enhanced attention to distinguishable and target-correlated spatial regions. These two mechanisms work in a complementary manner, allowing the network to capture spectral information while preserving spatial structural integrity and saliency. Consequently, more discriminative and expressive feature representations are achieved. The process can be expressed as follows:

$$\hat{f}_{out} = S_{ig}(PC_{onv}(C_{at}(\hat{Q}, \hat{K}))) \odot \hat{f}_{att} + \hat{f}_{att} \tag{27}$$

where $C_{at}(\cdot)$ denotes the concatenation function, $PC_{onv}(\cdot)$ represents the point-wise convolution operation, $S_{ig}(\cdot)$ indicates the Sigmoid activation function, and $\odot$ represents element-wise multiplication. $\hat{f}_{out} \in R^{H\times W\times B}$ denotes the output feature map of Spe-RWA.

E. Frequency-Domain Enhanced Feature Interaction Module

The FEFI module mainly consists of three key components: the multi-scale multi-strategy window transformer (MS2WT), the FDSE module, and the FDCE module, as illustrated in Fig. 1. Through the HSI-FEMM, MSI-FEMM, and DDI modules, FEM2Transformer performs two-stage fine-grained feature extraction and deep integration, obtaining high-quality representations of HSI and MSI. To further improve fusion performance, feature interaction between the two branches is introduced. Feature interaction in multi-source images is crucial for fusion tasks. In this work, the MS2WT module is designed using different-scale rectangular and triangular window transformers, as illustrated in Fig. 5(a). The MS2WT framework contains four sub-modules: Spa-RWT, Spe-RWT, spatial triangle window transformer (Spa-TWT), and spectral triangle window transformer (Spe-TWT). By jointly introducing two windowing strategies and multiple window scales, the MS2WT module enables more sufficient and diverse interaction and complementarity across the spatial and spectral domains of HSI and MSI, thereby effectively enhancing fusion capability. As described in the previous sections on FDSE and FDCE, the FDCE module enhances key spectral channel information in HSI through frequency-domain feature extraction, while the FDSE module enhances spatial detail information in MSI via frequency-domain modeling. Therefore, both FDSE and FDCE improve the frequency-domain

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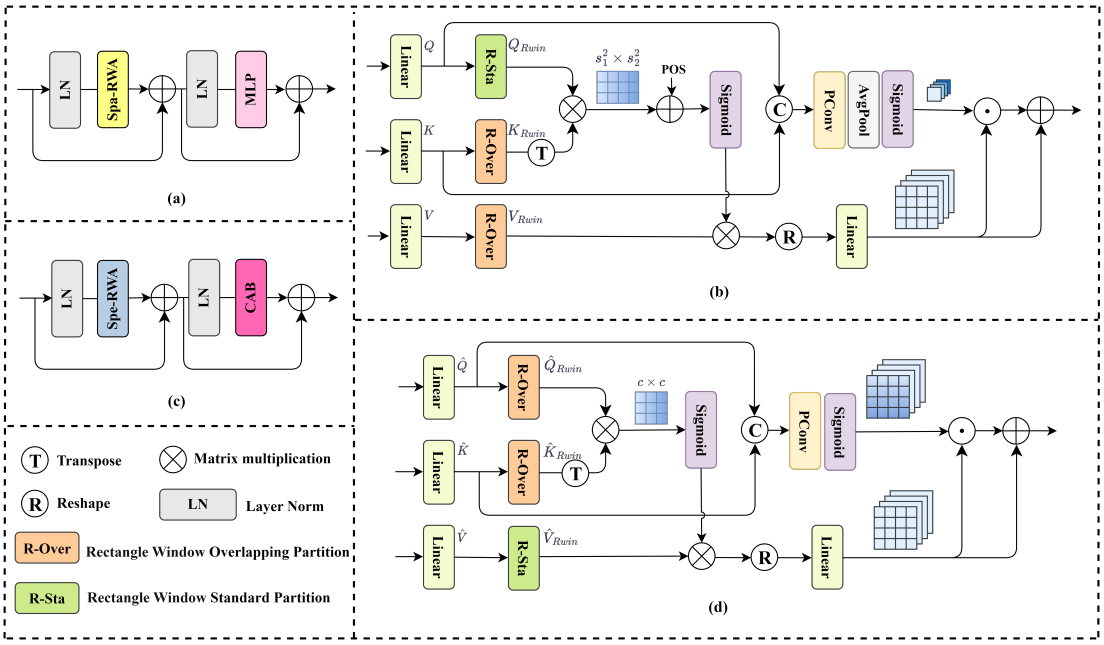


Fig. 4. Network structure of the main components in the DDI module. (a) Network structure of Spa-RWT, (b) Network structure of Spa-RWA, (c) Network structure of Spe-RWT, (d) Network structure of Spe-RWA.

representation capability of feature maps. By integrating these two frequency-enhanced modules with the MS2WT module, the FEFI module performs frequency-aware feature modeling from a spatial–spectral dual-domain collaborative perspective, achieving complementary enhancement of HSI and MSI features and further improving both spatial and frequency representation capabilities. The detailed computational steps of the FEFI module are listed below:

$$F_{hm}^{3,0} = M_{S2WT}(F_{h,m}^{2}) \tag{28}$$

$$F_{hm}^{3} = F_{DSE}(F_{hm}^{3,0}) + F_{DCE}(F_{hm}^{3,0}) + F_{hm}^{3,0} \tag{29}$$

where $M_{S2WT}(\cdot)$ denotes the mapping function of the MS2WT module, $F_{DSE}(\cdot)$ represents the mapping function of the FDSE module, and $F_{DCE}(\cdot)$ represents the mapping function of the FDCE module. $F_{hm}^{3,0} \in \mathbb{R}^{H\times W\times B}$ denotes the output feature map of the MS2WT module, and $F_{hm}^{3} \in \mathbb{R}^{H\times W\times B}$ denotes the output feature map of the FEFI module.

Next, we give a comprehensive introduction to the MS2WT module. Within the MS2WT module, the spatial window transformers (including Spa-RWT and Spa-TWT) and spectral window transformers (including Spe-RWT and Spe-TWT) are responsible for complementary feature interaction between different modalities. Specifically, the spatial window transformer focuses on enhancing MSI features using abundant spectral representations provided by HSI. Its mechanism constructs a cross-modal attention structure along the spatial dimension, where the key and value vectors are derived from MSI feature maps, while the query vector comes from HSI feature maps, thereby enabling HSI-guided enhancement of MSI representations. Correspondingly, the spectral window transformer aims to complement HSI by leveraging the strong spatial representation capability of MSI. The cross-attention mechanism is conducted along the spectral dimension, where the key and value vectors are obtained from HSI feature maps, while the query vector is derived from MSI feature maps, thereby enhancing the spatial detail representation of HSI. It is important to emphasize that the interaction between these two types of transformers follows a sequential order. First, the MS2WT module uses the spatial window transformer to guide MSI feature modeling with spectral characteristics from HSI. Subsequently, it adopts the spectral window transformer to further refine HSI spatial structures using the enhanced MSI features. This process is performed in a layer-wise alternating manner within the network, enabling progressive integration of spatial and spectral characteristics across the two modalities, and ultimately achieving deep interaction and mutual complementarity between HSI and MSI features.

In the MS2WT module, the two cross-attention branches are designed to operate in opposite directions and are applied in an alternating, layer-wise manner. In the spatial branch (Spa-RWT and Spa-TWT), the query is derived from HSI while the key and value are derived from MSI, so that the rich spectral representations of HSI are injected into MSI feature learning (HSI $\rightarrow$ MSI). In the spectral branch (Spe-RWT and Spe-TWT), the query is derived from MSI while the key and value are derived from HSI, so that the strong spatial representations of MSI are used to refine HSI features (MSI $\rightarrow$ HSI). Because the spatial branch is applied first and its enhanced MSI output is then fed into the spectral branch in the next round, the two branches form a progressive loop in which HSI and MSI continuously refine each other, constituting a genuine bidirectional information exchange rather than a unidirectional transfer. The MS2WT module performs a total of four groups of feature interaction operations. Among them, the interaction processes based on Spa-RWT and Spe-RWT are conducted three times in a progressive multi-stage manner to achieve coarse-to-fine fusion. The interaction based on Spa-TWT and Spe-TWT is applied in the final stage to further

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enhance the depth and breadth of feature fusion, ensuring effective spatial–spectral complementarity in the final representation. The MS2WT module simultaneously adopts two different window partitioning strategies: rectangular windows and triangular windows. The rectangular window introduces an overlapping mechanism and applies self-attention within each window to capture local spatial dependencies. Meanwhile, the overlapping regions facilitate information propagation across windows, enabling long-range feature interactions and global context modeling. In contrast, the triangular window [30] defines local regions along fixed directional layouts and focuses on direction-sensitive structural features within each window. Although its receptive field is limited, it has inherent advantages in modeling directional structures and edge details, effectively compensating for the directional perception limitations of rectangular windows. Moreover, due to its non-equidistant geometric structure, the triangular window allows more non-redundant translation patterns compared to rectangular windows, which helps reduce boundary artifacts, improve robustness to non-centered image patterns, and enhance directional sensitivity and fine-grained spatial representation. By jointly leveraging both rectangular and triangular window-based attention mechanisms, the MS2WT module preserves the advantages of rectangular windows in global context modeling while incorporating the directional sensitivity and boundary awareness of triangular windows. This effectively reduces structural bias arising from single-window strategies, boosting the model's sophisticated spatial modeling performance while optimizing reconstruction accuracy.

The two rectangular-window transformers (including Spa-RWT and Spe-RWT) have already been introduced in the DDI module. In that context, they adopt a self-attention mechanism to comprehensively explore and integrate the intrinsic spatial and spectral characteristics of HSI and MSI, respectively. In the FEFI module, however, these two rectangular-window transformers employ a cross-attention structure for information exchange and complementary learning across different modalities, thereby further enhancing fusion performance. The two triangular-window transformers (including Spa-TWT and Spe-TWT) are presented in Fig. 5(b) and Fig. 5(c), respectively. The Spa-TWT module mainly utilizes spatial triangle window attention (Spa-TWA) to complement MSI using the abundant spectral information from HSI, and its architecture is illustrated in Fig. 5(d). The Spe-TWT module mainly employs spectral triangle window attention (Spe-TWA) to complement HSI using the rich spatial information from MSI, and its architecture is shown in Fig. 5(e). For the three groups of rectangular-window transformers, the window sizes are different. In the first and second groups, the standard window size is $s_1 = 4$, and the overlapping window size is $s_2 = 6$, with a window stride of 4. In the third group, the standard window size is $s_1 = 8$, and the overlapping window size is $s_2 = 12$, with a window stride of 8. As for the triangular-window transformers, the window size is set to $s_3 = 16$. By introducing multi-scale window configurations, the MS2WT module is able to progressively extract feature representations from fine-grained to coarse-grained levels. This not only strengthens the model's capability to perceive complicated structures (e.g., edges and texture variations) and dynamic changes (e.g., scale variations), but also improves its capability to model relationships between spatial and spectral information across different scales, thereby comprehensively strengthening the expressive power of fused features. The detailed computation process of the MS2WT module is presented as follows:

$$\begin{cases} F_m^{2,i+1} = Spa_{RWT}(F_m^{2,i}, F_h^{2,i}) \\ F_h^{2,i+1} = Spe_{RWT}(F_m^{2,i+1}, F_h^{2,i}) \end{cases}, \ i = 0, 1, 2 \tag{30}$$

$$\begin{cases} F_m^{2,4} = Spa_{TWT}(F_m^{2,3}, F_h^{2,3}) \\ F_h^{2,4} = Spe_{TWT}(F_m^{2,4}, F_h^{2,3}) \end{cases} \tag{31}$$

where $Spa_{TWT}(\cdot)$ denotes the mapping function of the Spa-TWT module, and $Spe_{TWT}(\cdot)$ represents the mapping function of the Spe-TWT module. $F_m^{2,0}$ and $F_h^{2,0}$ denote the input feature maps, respectively. $F_m^{2,i}$ and $F_h^{2,i}$ represent the MSI feature map after i-round spectral supplementation operations and the HSI feature map after i-round spatial supplementation operations, respectively. Finally, the MS2WT module passes the fully interacted and complemented feature maps through convolution layers, residual skip connections, and linear normalization operations to generate the final feature representations. The detailed process is as follows:

$$\begin{cases} F_m^{3,0} = L_n(C_{onv}(F_m^{2,4}) + F_m^{2,0}) \\ F_h^{3,0} = L_n(C_{onv}(F_h^{2,4}) + F_h^{2,0}) \end{cases} \tag{32}$$

where $F_m^{3,0}$ and $F_h^{3,0}$ denote the output feature maps of the MS2WT module.

F. Loss Function

To enhance the model's training stability and generalization capability, we incorporate a combined constraint for spatial-domain loss L_s and frequency-domain loss L_f in the loss function design, aiming to enhance image fusion performance from different perspectives. L_s quantifies the per-pixel deviation between network prediction and reference ground truth (GT), which helps preserve structural details and local textures and improves the perceptual quality of the fused image. L_f, on the other hand, implements frequency-domain conversion based on the Fourier transform and optimizes spectral characteristics, empowering the model to effectively capture global contextual information. The two types of losses jointly guide the optimization process during training, enabling the fused image to achieve superior performance in both spatial consistency and frequency-domain representation. The comprehensive loss function is established in the following form:

$$L = L_s + \lambda L_f \tag{33}$$

where,

$$L_s = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{Z}_i - Z_i \right\|_1 \tag{34}$$

$$L_f = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathcal{F}(\hat{Z}_i) - \mathcal{F}(Z_i) \right\|_1 \tag{35}$$

where $\hat{Z}_i$ denotes the model's predicted fused image, Z_i denotes the GT, $\mathcal{F}(\cdot)$ represents the Fourier transform, and λ is a balancing parameter, which is set to 0.1 in this paper.

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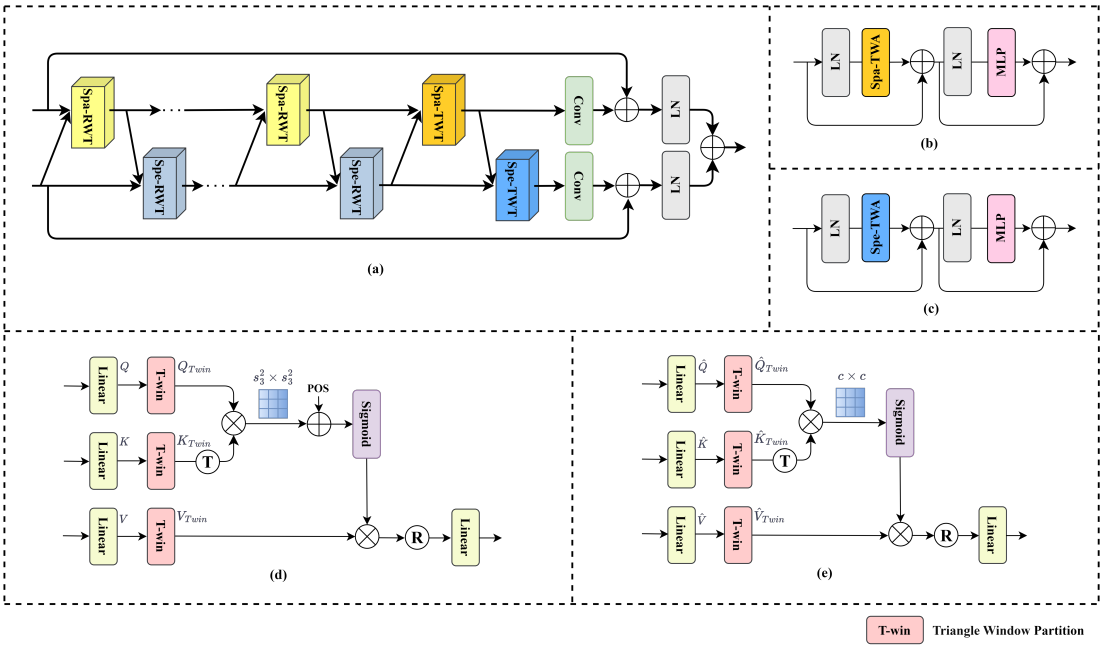


Fig. 5. Network structure of the MS2WT module and its main components. (a) Network structure of the MS2WT module, (b) Network structure of Spa-TWT, (c) Network structure of Spe-TWT, (d) Network structure of Spa-TWA, (e) Network structure of Spe-TWA.

IV. Experimental Validation and Result Analysis

In this section, we elaborate on the experimental parameter arrangement and performance assessment of the novel framework. We first describe the public datasets adopted for model training and testing. Next, we give the detailed experimental specifications. Thereafter, we exhibit experimental results and comparative analyses between the developed method and cutting-edge algorithms. Furthermore, ablation tests are carried out to confirm the practical effectiveness of every functional component in the established model. Lastly, the computational cost of the proposed network is analyzed and benchmarked against other existing mainstream methods.

A. Datasets

The reliability of the designed architecture is validated utilizing three prevalent simulated HSI databases, namely CAVE [31], Chikusei [32], and Houston [33]. The CAVE database consists of 32 indoor hyperspectral scenes. In this research, 20 images are picked randomly for model training, and the leftover 12 samples serve as test data. The Chikusei dataset possesses 128 spectral bands. During the preprocessing stage, 18 noisy channels are removed, and the remaining 110 channels are retained for fusion experiments. Subsequently, a central area of size 2048×2048 is cropped and divided into 16 sub-images with dimensions of 512×512×110. Among them, 10 patches are randomly employed for training, and the other 6 patches are used for evaluation. The Houston dataset contains 48 spectral bands. Two bands with inferior signal-to-noise ratios are excluded in advance, and 46 valid bands are retained for subsequent experimental analyses. The complete dataset is partitioned into a 2×7 grid structure, and the 512×512 area at the upper-left corner of each grid is selected as the GT patch. Seven patches are randomly sampled for model training, whereas the other seven patches are employed for testing.

As the proposed network adopts a supervised training framework, HR-HSIs are essential to serve as GT. However, authentic GT data for real-world scenarios is generally unavailable. For this reason, the original HSIs are taken as the high-resolution reference. Specifically, the original HSI data is preprocessed via a 5×5 Gaussian filter with a standard deviation of 2, followed by downsampling according to a fixed scaling factor to produce LR-HSIs. Meanwhile, HR-MSIs are generated by the corresponding spectral response function on the HR-HSI data. In terms of spectral response configuration, the CAVE dataset utilizes the 31×3 spectral response matrix of the Nikon D700 camera, while the Chikusei and Houston datasets adopt WorldView-2 spectral response functions with matrix dimensions of 110×8 and 46×3, respectively. To strengthen the model's generalization performance and enlarge the training dataset scale, 64×64 image patches are randomly cropped from the original HR-HSI. The paired LR-HSI and HR-MSI samples are then synthesized through the above degradation pipeline. Furthermore, common data augmentation strategies including image rotation and flipping are applied to improve the diversity of training samples.

B. Experimental Settings

To quantitatively validate the superior performance of the constructed framework, nine objective evaluation indicators consistent with the evaluation system in [21] are adopted to assess the comprehensive performance of different fusion algorithms. The adopted metric system includes peak signal-to-noise ratio (PSNR), root mean square error (RMSE), error relative global dimensionless synthesis (ERGAS), spectral angle mapper (SAM), universal image quality index (UIQI), structural similarity index measure (SSIM), mean absolute error (MoAE), correlation coefficient (CC), and Q2n. Among them, Q2n acts as a multi-band extended indicator derived

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from the universal image quality index. To further validate the efficacy of the developed framework, comprehensive comparative experiments are performed with four traditional HMIF algorithms and seven advanced deep learning-based HMIF methods. The selected traditional HMIF techniques are NLSTF [34], LTTR [35], CNMF [36], and CTDF [37]. Meanwhile, the deep learning-based comparison methods consist of MIMO-SST [38], DSPNet [39], AMSF [40], SST [41], SMGU-Net [6], MCIFNet [42], and DPFormer [21].

The devised framework is developed on the basis of Python 3.7 and PyTorch 1.12.1, and the entire training pipeline is executed on a workstation equipped with the Ubuntu 22.04 operating system. The graphics card is NVIDIA RTX A4000. In the training phase, the Adam optimizer is adopted with a batch size of 32. For the CAVE and Chikusei datasets, the initial learning rate is designated as 0.0008, and the network is trained for 300 epochs. For the Houston dataset, the initial learning rate is configured to 0.0003 with 160 training epochs. To accelerate the convergence of the model, a learning rate decay mechanism is introduced. Concretely, during the first 149 epochs, the learning rate is multiplied by a factor of 0.95 every 5 epochs, enabling a gradual decline of the learning rate during the network optimization.

To ensure a fair and rigorous comparison, we retrain all comparative methods from scratch using their respective official codes/configurations under the same experimental conditions as our method. Each method retains only its officially recommended hyperparameter configuration, yet all methods share a unified data preparation, partitioning strategy, and assessment protocol. Therefore, the quantitative results of all comparative methods as well as our method are produced under completely identical conditions, ensuring that the metrics are directly comparable and the comparison is fair.

C. Experimental Results on Simulated Datasets

The proposed model undergoes performance assessment over three simulated databases, namely CAVE, Chikusei, and Houston. Nine quantitative evaluation indices are utilized to conduct an all-around assessment of image fusion quality. Numerical comparison outcomes of diverse HMIF methods are summarized in Table I, Table II, and Table III, where all listed values stand for the average measurement calculated from all test samples. Marks highlighted in red represent the best achievement, while blue ones denote the suboptimal outcome. In addition, single-band error diagrams are visualized to offer straightforward performance observation. Spectral profiles extracted from typical local areas are also plotted so as to examine how well the restored images maintain spectral characteristics consistent with GT data.

1) Experimental Results on the CAVE Dataset: Table I reports the quantitative comparison results of different fusion approaches on the CAVE testing dataset. The second row indicates the best values for each metric. The proposed method attains the optimal performance across all evaluation indicators. DPFormer achieves the second-best performance across all nine evaluation metrics. Among the conventional methods, LTTR shows relatively competitive performance, obtaining a PSNR value of 47.78 dB and even outperforming the deep learning-driven AMSF method on this metric; however, it remains inferior to AMSF on the other evaluation indicators. CNMF produces the highest SAM value, indicating obvious spectral distortion in the reconstructed results. NLSTF yields the lowest performance on all metrics except SAM, suggesting that it has the weakest capability in preserving image details.

To visually assess the reconstruction capability of different algorithms, a test image named "fake_and_real_food" is randomly selected. The residual maps between the reconstructed outputs and the GT at the 29th spectral band are displayed, and several representative local areas are further enlarged, as illustrated in Fig. 6. According to Fig. 6, the proposed framework achieves the minimum overall reconstruction error. In comparison with deep learning-driven approaches, conventional algorithms generally show relatively poorer performance. Besides the enlarged regions, obvious reconstruction errors can also be observed in the upper area of the image (e.g., the paint palette region), indicating that traditional algorithms have difficulty preserving abundant MSI information because of their insufficient capability in representing complicated patterns and nonlinear correlations. Among the deep learning-based algorithms, AMSF exhibits the most noticeable reconstruction errors, whereas MIMO-SST, SST, and DPFormer achieve comparatively better results. However, from the enlarged strawberry region, it is evident that the proposed framework generates the fewest bright error artifacts. Overall, the proposed framework achieves the lowest reconstruction error in both high-frequency regions (e.g., edges and textures) and low-frequency areas.

To visually illustrate the capability of different fusion algorithms in maintaining spectral characteristics, the spectral curves corresponding to the fusion outputs produced by various algorithms for the "fake_and_real_food" image at pixel position (506, 301) are plotted, and several representative local regions are enlarged for comparison, as illustrated in Fig. 7. The spectral curve derived from the proposed framework is highly consistent with GT, demonstrating that the proposed framework possesses excellent spectral consistency.

2) Experimental Results on the Chikusei Dataset: Table II presents the objective scoring results of diverse fusion strategies on its test set. Statistical data reveal that our presented framework obtains top performance values for every assessment criterion. DPFormer gains optimal values in three metrics and suboptimal results in the rest six items, reflecting its powerful image restoration ability. Several deep learning models including MIMO-SST, SMGU-Net, and MCIFNet, achieve decent fusion quality, with partial indicators reaching the second rank. Within traditional algorithms, CTDF shows the most competitive performance. It achieves a PSNR of 54.90 dB, and its comprehensive effect even exceeds that of AMSF and DSPNet.

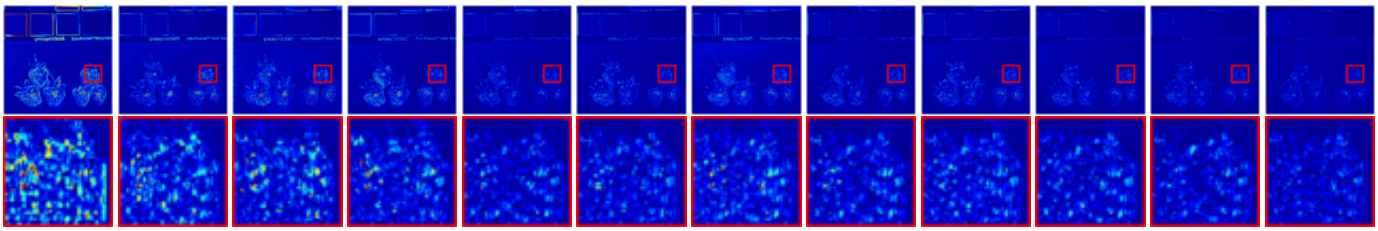
Similarly, the residual difference maps between reconstructed outputs and GT corresponding to the 18th spectral band for the "2_p_1_2" test image from the Chikusei dataset are plotted, and representative local regions are selected for magnification, as shown in Fig. 8. It is clearly observable that traditional algorithms (NLSTF, LTTR, and CNMF) and deep

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TABLE I
OBJECTIVE ASSESSMENT METRICS OF DIFFERENT FUSION METHODS ON THE CAVE TEST DATASET. BOLD VALUES INDICATE THE BEST RESULTS, UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS.

Methods	RMSE	PSNR	ERGAS	SAM	UIQI	SSIM	MoAE	CC	Q2n
NLSTF	2.9582	39.7532	3.6804	4.8524	0.9062	0.9788	1.3273	0.9931	0.8894
LTTR	1.3535	47.7806	1.6724	4.0357	0.9231	0.9918	0.5791	0.9980	0.9489
CNMF	1.8446	45.5802	2.1746	5.4246	0.8879	0.9869	0.8394	0.9966	0.9266
CTDF	1.5803	47.2696	1.9100	4.0822	0.9264	0.9904	0.6190	0.9972	0.9237
MIMO-SST	0.9872	50.1941	1.2267	3.3266	0.9380	0.9951	0.4215	0.9991	0.9593
DSPNet	1.0279	49.6762	1.2684	3.5446	0.9364	0.9948	0.4470	0.9990	0.9559
AMSF	1.2591	47.5752	1.5964	4.0939	0.9074	0.9916	0.6256	0.9986	0.9481
SST	0.8958	50.8615	1.0977	3.3528	0.9395	0.9957	0.3869	0.9992	0.9609
SMGU-Net	1.0277	49.6711	1.2669	3.5472	0.9357	0.9948	0.4526	0.9990	0.9579
MCIFNet	0.9895	49.7544	1.2744	3.3743	0.9338	0.9948	0.4355	0.9991	0.9583
DPFormer	0.8562	51.1065	1.0564	3.2357	0.9421	0.9960	0.3698	0.9993	0.9629
Ours	**0.8230**	**51.3975**	**1.0218**	**3.1101**	**0.9426**	**0.9963**	**0.3508**	**0.9994**	**0.9653**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 6. Error maps of the fused images generated by different algorithms and the GT at the 29th band for the "fake_and_real_food" in the CAVE test dataset. (a) NLSTF, (b) LTTR, (c) CNMF, (d) CTDF, (e) MIMO-SST, (f) DSPNet, (g) AMSF, (h) SST, (i) SMGU-Net, (j) MCIFNet, (k) DPFormer, and (l) Ours.

TABLE II
OBJECTIVE ASSESSMENT METRICS OF DIFFERENT FUSION METHODS ON THE CHIKUSEI TEST DATASET. BOLD VALUES INDICATE THE BEST RESULTS, UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS.

Method	RMSE	PSNR	ERGAS	SAM	UIQI	SSIM	MoAE	CC	Q2n
NLSTF	2.3535	42.0254	2.5842	1.9341	0.9768	0.9848	1.3545	0.9889	0.8780
LTTR	1.4599	50.3491	1.5344	2.0112	0.9853	0.9903	0.6301	0.9951	0.9135
CNMF	0.8304	50.6935	1.3515	1.2939	0.9859	0.9956	0.4557	0.9976	0.9437
CTDF	0.5859	54.9004	1.0046	0.9627	0.9921	0.9976	0.2411	0.9983	0.9727
MIMO-SST	0.4325	57.2132	0.6426	0.7381	0.9956	0.9986	0.1576	0.9993	0.9801
DSPNet	0.6016	53.6617	1.0258	0.9179	0.9930	0.9977	0.2378	0.9984	0.9683
AMSF	0.6876	52.1792	1.0677	1.2439	0.9908	0.9967	0.3452	0.9984	0.9641
SST	0.5010	55.4514	0.8248	0.8281	0.9945	0.9982	0.1969	0.9989	0.9443
SMGU-Net	0.4222	57.1852	0.6386	0.7387	0.9955	0.9986	0.1498	0.9993	0.9820
MCIFNet	0.4298	56.8208	0.6302	0.7444	0.9954	0.9986	0.1564	0.9993	0.9802
DPFormer	0.4021	57.5959	0.5791	0.6915	**0.9959**	**0.9987**	0.1422	**0.9994**	0.9834
Ours	**0.3996**	**57.6207**	**0.5764**	**0.6877**	**0.9959**	**0.9987**	**0.1407**	**0.9994**	**0.9836**

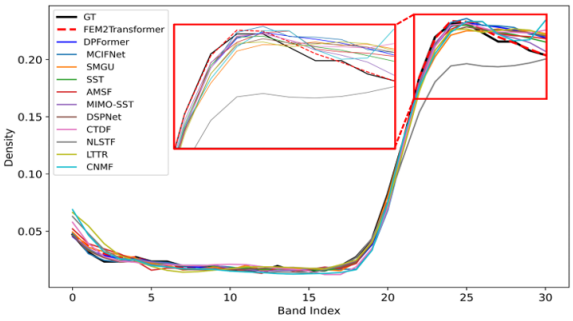


Fig. 7. Spectral curves at pixel (506, 301) of the fused images generated by different algorithms for the "fake_and_real_food" in the CAVE dataset.

learning-based algorithms (DSPNet, AMSF, and SST) exhibit relatively large fusion errors. The error maps of these methods display numerous and highly noticeable bright spots across the entire image, and the locally magnified maps show a large number of red areas, indicating substantial fusion errors. The remaining algorithms have relatively fewer bright spots across the entire image. However, comparison of the locally enlarged regions demonstrates that the proposed framework attains the strongest fusion capability, while DPFormer and CTDF also exhibit relatively low fusion errors.

To intuitively demonstrate the spectral preservation performance of different fusion algorithms, the spectral curves at pixel location (197, 261) for the "2_p_1_2" fused images obtained by different algorithms are plotted, and representative local regions are selected for magnification, as illustrated in Fig. 9. According to the full-band spectral curves in Fig. 9, the fused results generated by the traditional algorithms LTTR

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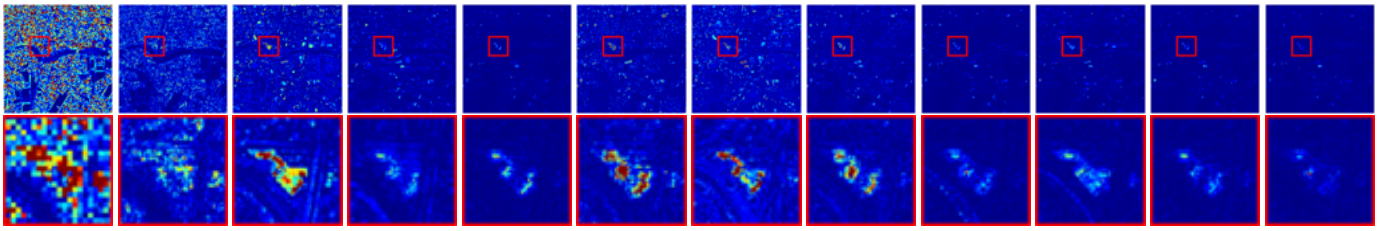


Fig. 8. Error maps of the fused images generated by different algorithms and the GT at the 18th band for the "2_p_1_2" in the Chikusei dataset. (a) NLSTF, (b) LTTR, (c) CNMF, (d) CTDF, (e) MIMO-SST, (f) DSPNet, (g) AMSF, (h) SST, (i) SMGU-Net, (j) MCIFNet, (k) DPFormer, and (l) Ours.

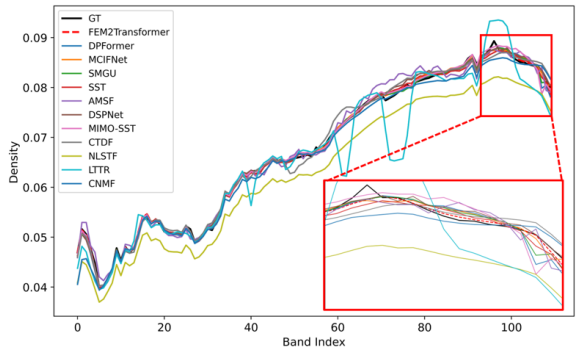


Fig. 9. Spectral curves at pixel (197, 261) of the fusion images yielded by each fusion algorithm for the "2_p_1_2" in the Chikusei dataset.

and NLSTF exhibit the largest deviations from the GT spectral curve. Furthermore, the locally enlarged regions indicate that the spectral curve corresponding to the fused result generated by the proposed framework most closely matches that of the GT, demonstrating the advantage of the proposed framework in preserving spectral information.

3) Experimental Results on the Houston Dataset: To quantitatively compare the performance of different fusion techniques, Table III lists average evaluation indicators for all comparative approaches on the Houston test dataset. The second row presents the optimal values of the evaluation metrics. As illustrated in the table, our method yields optimal values for eight evaluation indicators and achieves suboptimal performance on only one metric, with a negligible gap of merely 0.0001 compared with the best result. The deep learning-based DPFormer also presents competitive reconstruction performance, ranking second on seven metrics and attaining the best scores on two indices. In addition, SST and MCIFNet demonstrate satisfactory fusion ability, both obtaining the second-highest value in the CC indicator. For traditional HMIF algorithms, CTDF achieves the most promising fusion effect, while its overall performance is still substantially weaker than that of deep learning-based fusion methods.

Fig. 10 visualizes the error maps between the reconstructed images and the GT at the 37th spectral band for the "1_2" test image in the Houston dataset, and representative local regions are selected for magnification. As shown in Fig. 10, the fusion errors of the four traditional algorithms are all very prominent, with a relatively large number of red areas in the magnified regions. In contrast, the deep learning-based HMIF algorithms produce smaller fusion errors. Observing the locally magnified regions, among the deep learning-based algorithms, MIMO-SST and DSPNet exhibit poorer fusion results, whereas the proposed method attains the minimum fusion error.

Meanwhile, to intuitively analyze the spectral preservation capability of different fusion approaches, the spectral curves of the "1_5" reconstructed images in the Houston dataset at pixel location (470, 228) are plotted for different algorithms, and representative local regions are magnified, as shown in Fig. 11. As shown in Fig. 11, the full-band spectral curve comparisons reveal that the fused images produced by traditional algorithms LTTR, NLSTF, and CTDF still show obvious deviations from the GT, with the NLSTF fused image exhibiting the most significant deviation. The locally magnified results demonstrate that the fused result generated by the proposed framework exhibits the highest consistency with the GT spectral curve, which strongly confirms the superiority of the proposed framework in maintaining spectral information.

D. Ablation Study

To confirm the functional validity of each embedded module in the designed framework, ablation tests were carried out on the FEM2Transformer model based on the Houston dataset. The model consists of four main modules, each containing several sub-modules. The effectiveness of each main module and its sub-modules is analyzed separately, and the impact of the multi-strategy window and multi-scale window mechanisms on overall model performance is further investigated.

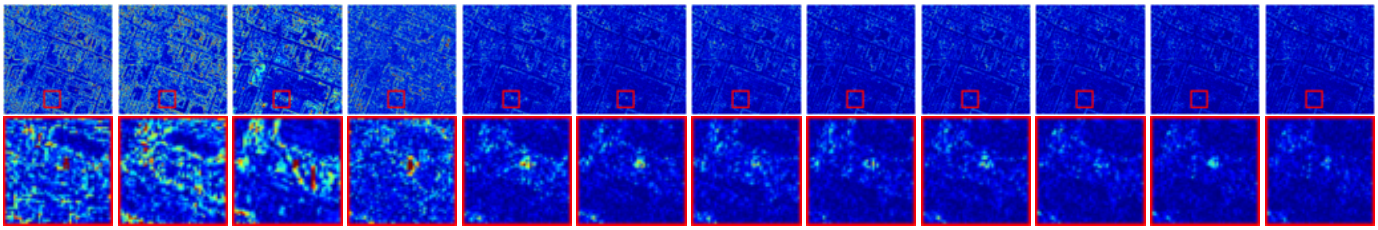
1) Ablation Study of Main Modules: To examine the functional validity of every module integrated into the designed framework, ablation studies were conducted on the HSI-FEMM module, the MSI-FEMM module, the DDI module, and the FEFI module. To demonstrate the contribution of the HSI-FEMM and MSI-FEMM modules to fine-grained feature extraction, the models without the HSI-FEMM module or the MSI-FEMM module are denoted as "w/o HSI-FEMM" and "w/o MSI-FEMM", respectively. To examine the contribution of the DDI module in comprehensively mining and integrating the corresponding spatial and spectral characteristics of HSI and MSI, the model without the DDI module is denoted as "w/o DDI". To validate that the FEFI module can effectively perform interactive fusion of the features extracted from the two branches, the model without the FEFI module is denoted as "w/o FEFI". The ablation results of different models are listed in Table IV, where the optimal values are highlighted in bold. As observed from Table IV, removing any individual module results in performance degradation across all metrics, which fully demonstrates that each module designed in this paper contributes significantly to HMIF. In particular, removing the FEFI module leads to the largest performance decline.

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TABLE III
OBJECTIVE QUANTITATIVE METRICS OF DIFFERENT FUSION METHODS ON THE HOUSTON TEST DATASET. BOLD VALUES INDICATE THE BEST RESULTS, UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS.

Methods	RMSE	PSNR	ERGAS	SAM	UIQI	SSIM	MoAE	CC	Q2n
NLSTF	6.1596	35.2398	2.3431	2.6378	0.9579	0.9150	3.5571	0.9823	0.5580
LTTR	5.9748	39.0728	2.1886	2.6896	0.9586	0.9026	3.3649	0.9825	0.5967
CNMF	5.5551	41.0498	1.9002	2.2632	0.9690	0.9429	2.8678	0.9863	0.5567
CTDF	4.8736	42.7747	1.6555	2.3700	0.9734	0.9370	2.4806	0.9898	0.6440
MIMO-SST	2.7481	45.9681	0.9381	1.5016	0.9912	0.9757	1.4301	0.9966	0.7410
DSPNet	2.3951	46.3900	0.8272	1.3633	0.9930	0.9817	1.2260	0.9974	0.7617
AMSF	2.6517	44.0756	0.9465	1.6117	0.9912	0.9786	1.4580	0.9969	0.7423
SST	2.1843	47.1524	0.7539	1.2844	0.9941	0.9840	1.1125	0.9979	0.7624
SMGU-Net	2.3556	46.4020	0.8144	1.3819	0.9930	0.9820	1.2108	0.9975	0.7573
MCIFNet	2.1753	46.7514	0.7562	1.3162	0.9939	0.9847	1.1272	0.9979	0.7634
DPFormer	1.9849	47.5892	0.6901	1.2404	0.9948	0.9863	1.0155	**0.9982**	**0.7810**
Ours	**1.9717**	**47.7078**	**0.6849**	**1.2242**	**0.9949**	**0.9864**	**1.0096**	**0.9982**	0.7809



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 10. Error maps of the fused images generated by different fusion algorithms and the GT at the 37th band for the "1_2" in the Houston dataset. (a) NLSTF, (b) LTTR, (c) CNMF, (d) CTDF, (e) MIMO-SST, (f) DSPNet, (g) AMSF, (h) SST, (i) SMGU-Net, (j) MCIFNet, (k) DPFormer, and (l) Ours.

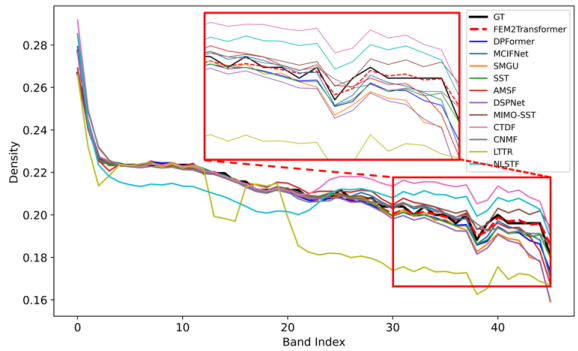


Fig. 11. Spectral curves at pixel (470, 228) of the fused images generated by different fusion algorithms for the "1_5" in the Houston dataset.

TABLE IV
ABLATION RESULTS OF DIFFERENT MODULES IN FEM2TRANSFORMER ON THE HOUSTON DATASET. BOLD VALUES INDICATE THE BEST RESULTS.

Ablation Model	RMSE	PSNR	ERGAS	SAM	UIQI	MoAE	Q2n
w/o HSI-FEMM	2.0424	47.5425	0.7067	1.2450	0.9945	1.0419	0.7751
w/o MSI-FEMM	2.0183	47.5232	0.7007	1.2424	0.9947	1.0334	0.7767
w/o DDI	2.0397	47.5217	0.7066	1.2453	0.9946	1.0422	0.7758
w/o FEFI	2.2917	46.7946	0.7909	1.3384	0.9936	1.1699	0.7640
Ours	**1.9717**	**47.7078**	**0.6849**	**1.2242**	**0.9949**	**1.0096**	**0.7809**

This indicates that sufficient interaction and fusion between the two modalities are highly crucial in HMIF tasks, while also confirming the capability of the FEFI module in promoting feature interaction and fusion.

2) Effectiveness Validation of Sub-modules: To evaluate the roles of the functional sub-modules contained within the main modules of FEM2Transformer, corresponding ablation studies were performed on these sub-modules. To examine the contribution of the SMP module to multi-scale spectral information extraction, the model without the SMP module is referred to as "w/o SMP". To demonstrate the contribution of the MSAX module to spatial multi-scale feature extraction and alignment, the model without the MSAX module is referred to as "w/o MSAX". To examine the contribution of the FDCE and FDSE modules to feature frequency-structure mining, the models without these two modules are denoted as "w/o FDCE" and "w/o FDSE", respectively. To demonstrate the contribution of the EP module to extracting spatial edge and texture details from MSI, the model without the EP module is denoted as "w/o EP". To evaluate the efficacy of the DDI module in dual-domain information integration, ablation studies were performed on the Spa-RWT and Spe-RWT sub-modules within the DDI module; the model without the Spa-RWT module (meaning that, within the DDI module, only spectral-domain features are enhanced) is denoted as "w/o Spa-RWT", and the model without the Spe-RWT module (meaning that, within the DDI module, only spatial-domain features are enhanced) is denoted as "w/o Spe-RWT". To examine the contribution of the MS2WT module to sufficient interactive fusion of the features extracted by the two branches, the model without the MS2WT module is referred to as "w/o MS2WT". The experimental findings of all ablation variants are presented in Table V, where bold text represents the optimal objective metric values. According to Table V, the comparison between the complete model and its ablation variants shows that removing any individual module causes different degrees of performance degradation on all objective assessment metrics. These results indicate that each sub-module serves an indispensable function within the entire framework and can improve feature representation and fusion capability from different aspects. The acquired results effectively validate the rationality of the

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TABLE V
ABLATION RESULTS OF THE SUBMODULES OF THE MAIN MODULES IN FEM2TRANSFORMER ON THE HOUSTON DATASET. BOLD VALUES INDICATE THE BEST RESULTS.

Ablation Model	RMSE	PSNR	ERGAS	SAM	UIQI	MoAE	Q2n
w/o SMP	2.0197	47.5551	0.7002	1.2375	0.9947	1.0321	0.7791
w/o MSAX	2.0498	47.4088	0.7112	1.2538	0.9944	1.0528	0.7735
w/o FDCE	2.0388	47.4892	0.7071	1.2384	0.9946	1.0378	0.7799
w/o EP	2.0341	47.5488	0.7045	1.2408	0.9946	1.0357	0.7798
w/o FDSE	2.0588	47.2795	0.7171	1.2455	0.9946	1.0461	0.7782
w/o Spa-RWT	2.0159	47.5492	0.6999	1.2412	0.9947	1.0309	0.7773
w/o Spe-RWT	2.0473	47.6023	0.7080	1.2312	0.9946	1.0360	0.7763
w/o MS2WT	2.2323	47.0537	0.7698	1.3148	0.9938	1.1352	0.7643
Ours	**1.9717**	**47.7078**	**0.6849**	**1.2242**	**0.9949**	**1.0096**	**0.7809**

TABLE VI
ABLATION RESULTS OF THE MULTI-STRATEGY WINDOW ON THE HOUSTON HYPERSPECTRAL DATASET. BOLD VALUES INDICATE THE BEST RESULTS.

Ablation Model	RMSE	PSNR	ERGAS	SAM	UIQI	MoAE	Q2n
RWT	2.0876	47.4482	0.7220	1.2476	0.9945	1.0595	0.7735
TWT	2.1442	47.3358	0.7398	1.2537	0.9942	1.0839	0.7718
Ours	**1.9717**	**47.7078**	**0.6849**	**1.2242**	**0.9949**	**1.0096**	**0.7809**

proposed structural model and highlight the necessity of inter-module collaborative operation.

3) Effectiveness Validation of the Multi-Strategy Window: To verify the effectiveness of jointly utilizing different window partitioning strategies in the MS2WT module, corresponding ablation experiments focused on the multi-window design strategy were conducted. Specifically, the triangular window transformers (including Spa-TWT and Spe-TWT) in the MS2WT module were uniformly replaced with rectangular window transformers (corresponding to Spa-RWT and Spe-RWT) to construct a model employing solely the rectangular window strategy, denoted as "RWT". Similarly, the rectangular window transformers (including Spa-RWT and Spe-RWT) in the MS2WT module were uniformly replaced with triangular window transformers (corresponding to Spa-TWT and Spe-TWT) to obtain a model employing solely the triangular window strategy, denoted as "TWT". Through these comparative experiments, the influence of different window partitioning strategies on overall model capability is systematically investigated. The experimental findings are presented in Table VI, where bold values denote the optimal objective metrics. As indicated in Table VI, the objective assessment results of both single-strategy window models are inferior to those of the multi-strategy window framework adopted in this paper, which fully verifies the effectiveness of the proposed multi-strategy window mechanism.

Table VI further indicates that the gain comes from the synergy of the two window types rather than from either mechanism alone. The multi-strategy model (Ours) consistently surpasses both the rectangular-only (RWT) and triangular-only (TWT) settings on every evaluation metric, while neither single-strategy variant reaches comparable performance. This complementarity is not a trivial transplantation of existing window mechanisms, but is instead motivated by the specific characteristics of HMIF. Rectangular overlapping windows excel at capturing long-range spatial context through an enlarged receptive field, which is particularly beneficial for enhancing the spatial detail of MSI. Triangular windows, with their direction-sensitive, non-equidistant geometry, are adept at modeling edges and oriented structures, which helps preserve the structural and directional fidelity of HSI. Since HMIF must simultaneously maintain spatial detail (dominated by MSI) and spectral fidelity (dominated by HSI), and the two modalities exhibit distinct statistical properties, the combination of rectangular and triangular windows therefore provides a task-aware, complementary modeling capacity that is uniquely advantageous for HMIF, as quantitatively confirmed by Table VI.

4) Visualization and Analysis of Cross-Attention Weights: To further verify the bidirectional HSI-MSI compensation mechanism of the MS2WT module beyond the quantitative ablation results above, we visualize the learned cross-attention weights of its multi-strategy window transformers. As illustrated in Fig. 12, the spatial branch (Spa-RWT and Spa-TWT) takes HSI as the query and MSI as the key/value, while the spectral branch (Spe-RWT and Spe-TWT) takes MSI as the query and HSI as the key/value. From Fig. 12(a) and (b), each HSI query position (each tile in the layout) exhibits a distinct and concentrated attention pattern over the MSI key region, and the center and corner windows display clearly different patterns, demonstrating that the spatial branch learns query-specific, spatially-adaptive HSI-MSI attention rather than uniform weighting. This contrast stems from the overlapping rectangular partition, which tiles the feature map into windows at distinct spatial positions so that the model adapts its HSI-MSI attention to each location rather than applying a position-invariant weighting. Fig. 12(c) shows that the triangular-window attention is concentrated along the diagonal with structured off-diagonal responses, confirming that the triangular window captures direction-sensitive spatial structures. This diagonal concentration arises because the triangular window partitions each region along a non-equidistant, directional layout: spatially adjacent tokens are aligned along the diagonal, while its directional edges yield the off-diagonal bands, manifesting direction-sensitive modeling. Fig. 12(d) and (e) reveal that the spectral branch builds strong feature-dimensional affinity between the two modalities. Since the spectral branch performs cross-attention along the spectral dimension, these matrices expose which feature components of the two modalities interact most strongly, directly evidencing the MSI and HSI feature compensation. Taken together, the attention visualizations empirically confirm that the multi-strategy window transformer establishes an effective, bidirectional information flow between HSI and MSI, in which the two modalities progressively refine each other, thereby substantiating the compensation mechanism described above.

5) Effectiveness Validation of the Multi-Scale Window: To validate the effectiveness of integrating windows with different scales within the MS2WT module, corresponding ablation experiments were conducted, focusing on various configuration schemes for the multi-scale windows. As discussed earlier, the MS2WT module employs three groups of rectangular window transformers with window sizes of 4, 4, and 8, respectively, followed by one group of triangular window transformers with a window size of 16. In this ablation study, multiple sets of different size combinations were established to assess their impact on the model. For these tests, the rectangular window sizes were chosen from $\{4, 8\}$, and the triangular window sizes were chosen from $\{4, 8, 16, 32\}$. Specifically, a model with

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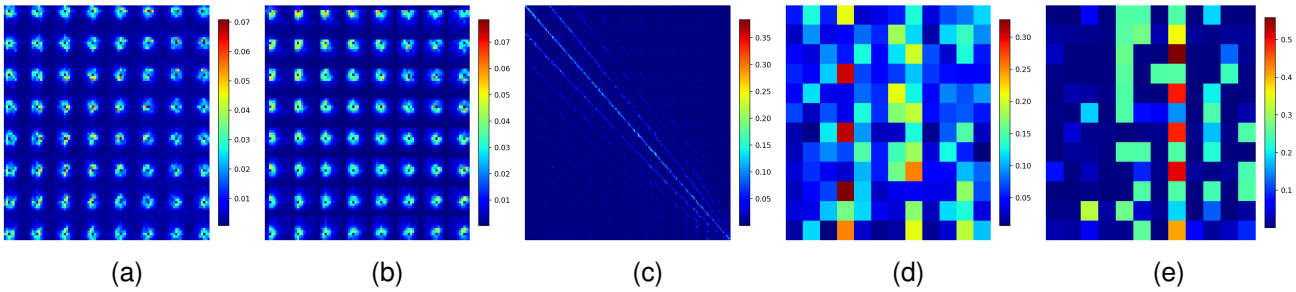


Fig. 12. Visualization of the learned cross-attention weights in the MS2WT module. (a) and (b) Spa-RWT spatial attention layouts for a center and a corner window. (c) Spa-TWT within triangular-window attention. (d) and (e) Feature affinity of the spectral branch (Spe-RWT and Spe-TWT).

TABLE VII
ABLATION RESULTS OF THE MULTI-SCALE WINDOW ON THE HOUSTON HYPERSPECTRAL DATASET. BOLD VALUES INDICATE THE BEST RESULTS.

Ablation Model	RMSE	PSNR	ERGAS	SAM	UIQI	MoAE	Q2n
448_4	2.0283	47.5375	0.7031	1.2496	0.9947	1.0369	0.7754
448_8	2.0127	47.5846	0.6982	1.2380	0.9948	1.0285	0.7786
448_32	1.9961	47.6168	0.6928	1.2284	0.9948	1.0209	0.7795
444_16	1.9936	47.6162	0.6918	1.2308	0.9947	1.0196	0.7797
888_16	2.0290	47.5439	0.7035	1.2378	0.9947	1.0366	0.7772
484_16	1.9877	47.6800	0.6897	1.2261	**0.9949**	1.0148	0.7801
844_16	2.0203	47.5968	0.7001	1.2350	0.9947	1.0302	0.7793
Ours	**1.9717**	**47.7078**	**0.6849**	**1.2242**	**0.9949**	**1.0096**	**0.7809**

TABLE VIII
ABLATION RESULTS OF THE LOSS FUNCTION WEIGHT ON THE HOUSTON HYPERSPECTRAL DATASET. BOLD VALUES INDICATE THE BEST RESULTS.

λ	RMSE	PSNR	ERGAS	SAM	UIQI	MoAE	Q2n
1	2.0332	47.5732	0.7041	1.2368	0.9947	1.0349	0.7778
0.1	**1.9717**	**47.7078**	**0.6849**	**1.2242**	**0.9949**	**1.0096**	**0.7809**
0.01	1.9948	47.5974	0.6931	1.2333	0.9948	1.0211	0.7801
0	2.0144	47.5132	0.7000	1.2381	0.9947	1.0291	0.7779

the three groups of rectangular window sizes set to 4, 4, and 8, and the last group's triangular window size set to 4 is denoted as "448_4", with other comparison models named according to the same convention. The experimental findings are presented in Table VII, where bold text indicates the optimal objective metric values. As Table VII shows, the fusion performance is sensitive to how the window scales are combined. The variant without small (size-4) rectangular windows, i.e., "888_16", yields the worst results on the RMSE and ERGAS metrics among the compared configurations, suggesting that small windows contribute to local-detail preservation. Among all tested combinations, our configuration "448_16" (with a triangular window size of 16) attains the best fusion performance and also outperforms the competitive setting "484_16". We therefore set the triangular window size to 16 in this study.

6) Effectiveness Validation of the Loss Function: To validate the contribution of the introduced frequency-domain loss, ablation experiments were conducted by varying the balancing weight λ in the total loss $L = L_s + \lambda L_f$, where L_f denotes the frequency-domain loss. Four settings were tested with $\lambda \in \{1, 0.1, 0.01, 0\}$, where $\lambda = 0$ corresponds to removing the frequency-domain loss entirely. The corresponding results are reported in Table VIII, where bold values indicate the optimal objective metrics. As Table VIII shows, removing the frequency-domain loss ($\lambda = 0$) or assigning it an inappropriate weight consistently degrades the fusion quality across all evaluation metrics, while the setting $\lambda = 0.1$ achieves the best performance and is therefore adopted in this work.

TABLE IX
INDICATORS OF COMPLEXITY FOR DIFFERENT ALGORITHMS ON THE CAVE DATABASE. BOLD VALUES INDICATE THE BEST RESULTS.

Method	Parameters (M)	FLOPs (G)	PSNR	SAM
MIMO-SST	1.6218	**98.1653**	50.1941	3.3266
DSP-Net	6.0551	422.1895	49.6762	3.5446
AMSF	1.4565	159.2093	47.5752	4.0939
SST	4.5772	936.9030	50.8615	3.3528
SMGU	0.7594	381.0955	49.6711	3.5472
MCIFNet	**0.5110**	146.7975	49.7544	3.3743
DPFormer	1.3696	159.3056	51.1065	3.2357
Ours	3.1263	286.2632	**51.3975**	**3.1101**

E. Model Complexity Analysis

To thoroughly examine the computational burden of the proposed model, comparisons of parameter scale and computational complexity among multiple deep learning fusion methods are carried out for the CAVE dataset. Table IX presents detailed parameter volume and FLOPs of each HMIF technique, where bold content denotes the best results. As indicated in Table IX, the parameter count of the proposed method is smaller than that of DSP-Net and SST, but higher than that of DPFormer, MCIFNet, AMSF, MIMO-SST, and SMGU. In terms of computational complexity, its FLOPs are lower than those of DSP-Net, SST, and SMGU, but higher than those of DPFormer, MCIFNet, AMSF, and MIMO-SST. Although the parameter counts and FLOPs of the proposed algorithm are at an intermediate level compared with the aforementioned methods, the proposed algorithm achieves significantly superior performance on objective fusion metrics, realizing an improved balance between performance and efficiency. Moreover, combining the ablation results from Tables IV and V reveals that the parameter counts and FLOPs within the proposed model are primarily concentrated in the MSAX and FEFI modules. Specifically, the MSAX module extracts feature maps at three different scales and is simultaneously applied in both the HSI-FEMM and MSI-FEMM modules, thus introducing a relatively large number of parameters. Meanwhile, the FEFI module incorporates frequency-domain enhancement operations on top of sufficient interactive fusion of the dual-branch features; this entire process involves dense computations, thereby resulting in higher FLOPs. Overall, although certain modules increase the model's parameter count and computational complexity, these designs effectively improve the network's feature representation and fusion capabilities, significantly enhancing the final fusion performance.

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TABLE X
NO-REFERENCE QUALITY METRICS OF DIFFERENT FUSION ALGORITHMS ON THE WV-3 DATASET. BOLD VALUES INDICATE THE BEST RESULTS, UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS. FOR BRISQUE AND D_λ, LOWER VALUES INDICATE BETTER QUALITY, WHEREAS FOR QNR, HIGHER VALUES INDICATE BETTER QUALITY.

Method	CNMF	DSPNet	MIMO-SST	AMSF	SST	SMGU	MCIFNet	DPFormer	Ours
BRISQUE	75.6576	45.7536	51.8472	67.0082	66.4519	45.3788	<u>43.9416</u>	44.4103	**43.8797**
QNR	0.8798	0.8996	**0.9261**	<u>0.9230</u>	0.8759	0.8842	0.9132	0.9175	0.9211
D_λ	0.0270	<u>0.0173</u>	0.0176	0.0242	0.0178	0.0246	0.0224	0.0182	**0.0161**

V. EXPERIMENTS ON REAL-WORLD DATASET

To validate the practicability of the FEM2Transformer in actual application scenarios, additional experimental verification of our approach was carried out on the authentic remote sensing dataset WorldView-3 (WV3) [43], which spans the visible and near-infrared spectral ranges, containing PAN together with MSI (with 8 bands). During model training and testing, images in the dataset were cropped into several small patches. In the training set, the size of PAN patches is 64×64, and the size of LR-MSI patches is 16×16. The test set comprises 20 pairs of PAN images (size 512×512) and LR-MSI images (size 128×128). Since real-world datasets lack GT images, no-reference metrics are adopted for quantitatively assessing the fused images. Specifically, the blind/referenceless image spatial quality evaluator (BRISQUE) metric, the quality with no reference (QNR) and the spectral distortion (D_λ) were selected to assess the quality of the fused outcomes. BRISQUE [44] can effectively measure the extent of image distortion and provide reliable quality assessment results. QNR [45] provides an overall no-reference measure of fusion quality, whereas D_λ reflects the spectral distortion of the fused image. Since traditional fusion algorithms such as LTTR, NLSTF, and CTDF rely on explicit degradation models—which are non-existent for real-world datasets—the designed approach is benchmarked against alternative deep learning algorithms on the practical dataset to reflect its outstanding capability.

Table X presents the quantitative evaluations of various fusion algorithms on the real-world WV3 dataset. As shown in Table X, the proposed algorithm achieves the optimal BRISQUE score, while MCIFNet obtains the second-best value. In addition to the optimal BRISQUE score, the proposed method also achieves the best performance on D_λ and ranks third on QNR. Although the QNR result is not the best, the gap to the top-performing methods is negligible, demonstrating that our method maintains highly competitive no-reference fusion quality on real data and is by no means inferior. To visually demonstrate the fused images produced by different algorithms more objectively, one image (Image 20) was randomly selected from the WV3 test set. Pseudo-color images composed of the 1st, 3rd, and 5th bands are utilized to display the fusion outcomes of different algorithms, and a specific region (indicated by a red box) is enlarged for closer inspection, as presented in Fig. 13. As observed from Fig. 13, compared with the LR-MSI, the fused image produced by CNMF exhibits obvious color distortion in the roof regions. The fused image from MIMO-SST contains stripe noise, while the overall brightness of the AMSF fusion result is overly dark. The SST fused image suffers from noticeable edge blurring and insufficient detail recovery. Observing the locally magnified region of the SMGU fused image reveals that the red area of the roof appears lighter, indicating some color distortion. Meanwhile, the fusion results of DSPNet, MCIFNet, DPFormer, and our method are relatively close. In conclusion, our method demonstrates the most satisfactory fusion performance for the real-world dataset.

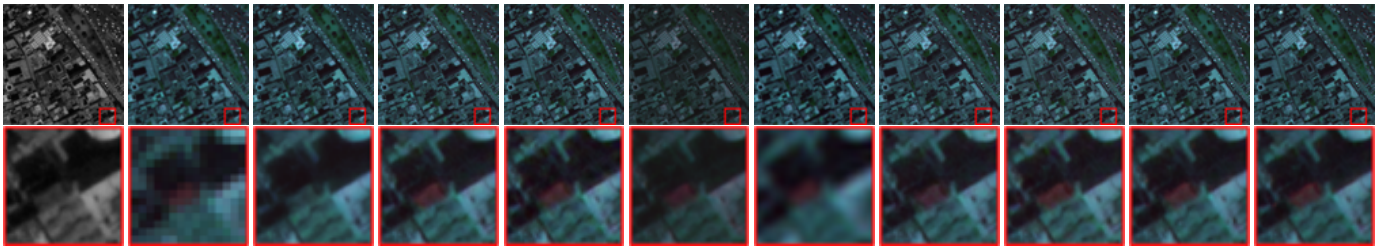
To intuitively evaluate the ability of different fusion algorithms to preserve spectral information during real-world dataset testing, a test image (Image 7) was randomly selected, and the spectral profiles of the fusion outcomes from various algorithms were plotted at pixel location (100, 112), as illustrated in Fig. 14. It can be found that compared with the spectral profile of LR-HSI, the fusion outcome acquired by our method demonstrates higher consistency in both spectral shape and amplitude, reflecting optimal spectral reconstruction performance. In contrast, CNMF exhibits significant spectral deviation, indicating severe spectral reconstruction errors. These comparisons thoroughly demonstrate that our method possesses a marked superiority in spectral information preservation.

VI. CONCLUSION

In this work, we proposed FEM2Transformer, a frequency-domain-enhanced multi-scale multi-strategy transformer for HMIF. By jointly exploiting spatial-spectral dual-domain frequency cues and multi-scale multi-strategy window transformers, the framework provides a more general and effective paradigm for fine-grained and complementary HMIF, overcoming the limited receptive field and single-domain modeling limitations of previous approaches. Empirical tests on three simulated datasets and one real-world dataset demonstrate that the proposed FEM2Transformer outperforms mainstream state-of-the-art methods regarding quantitative metrics and visual fusion effects. This further validates the functional effectiveness and environmental robustness of the developed framework in complex and diverse application environments. The consistent gains across both simulated and real-scene scenarios indicate that the incorporated frequency-domain enhancement and multi-strategy interaction generalize well beyond a single dataset, confirming the practical value of the proposed design. Although the FEM2Transformer achieves outstanding fusion performance, the scale of its model parameters is not entirely advantageous. This is primarily due to the MSAX module, which introduces a substantial number of convolutional parameters during the feature extraction process. This module requires the extraction of feature maps at multiple scales within both the HSI and MSI branches, and it employs downsampling convolutions with a high number of channels, thereby increasing the overall parameter count. To tackle this

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 13. Pseudo-color images composed of the 1st, 3rd, and 5th bands of the fusion results for Image 20 in the WV-3 dataset produced using different fusion methods. (a) PAN, (b) LR-HSI, (c) CNMF, (d) DSPNet, (e) MIMO-SST, (f) AMSF, (g) SST, (h) SMGU, (i) MCIFNet, (j) DPFormer, and (k) Ours.

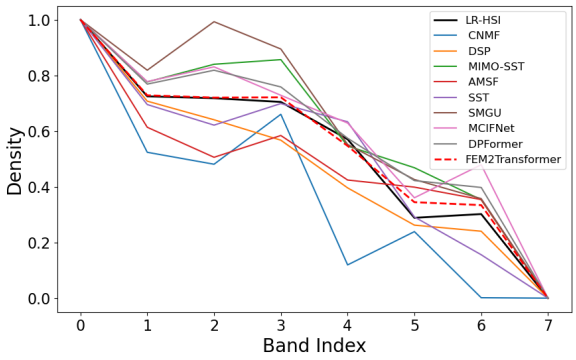


Fig. 14. Spectral curves at pixel (100, 112) of the fusion images generated by different fusion algorithms for Image 7 in the WV-3 dataset.

issue, future work will concentrate on structural optimization and lightweight design for the MSAX module while maintaining fusion performance, with the goal of further improving the practicality and computational efficiency of the framework. Extending the proposed frequency-aware multi-strategy fusion framework to other multi-source remote sensing fusion tasks also constitutes a promising direction for future research.

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